

Aurora kinase A promotes trained immunity via regulation of endogenous S-adenosylmethionine metabolism

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eLife Assessment

The authors use a range of techniques to examine the role of Aurora Kinase A (AurA) in trained immunity. The study is hypothesis driven, it uses **solid** experimental approaches, and the data are presented in a logical manner. The findings are **valuable** to the trained immunity field because they provide an in-depth look at a common inducer of trained immunity, beta-glucan.

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Abstract

Innate immune cells can acquire a memory phenotype, termed trained immunity, but the mechanism underlying the regulation of trained immunity remains largely elusive. Here, we demonstrate that inhibition of Aurora kinase A (AurA) dampens trained immunity induced by β -glucan. ATAC-seq and RNA-seq analysis reveals that AurA inhibition restricts chromatin accessibility of genes associated with inflammatory pathways such as JAK-STAT, TNF and NF- κ B pathways. Specifically, AurA inhibition promotes nuclear localization of FOXO3 and the expression of glycine N-methyltransferase (GNMT), a key enzyme responsible for S-adenosylmethionine (SAM) consumption. Metabolomic analysis confirms a reduction in SAM level upon AurA inhibition. As a result of SAM deficiency, trained mouse macrophages exhibit decreased H3K4me3 and H3K36me3 enrichment on gene regions of *Il6* and *Tnf*. Additionally,

the tumor inhibition effect of β -glucan is notably abolished by AurA inhibition. Together, our findings identify an essential role of AurA in regulating trained immunity via a methylation-dependent manner by maintaining endogenous SAM level through mTOR-FOXO3-GNMT axis.

Introduction

Trained immunity is depicted as a memory state of innate immune cell, independent on adaptive immunity (Naik & Fuchs, 2022 [↗](#); Ochando, Mulder, Madsen, Netea, & Duivenvoorden, 2023 [↗](#)). It can be induced by *Candida albicans*, Bacillus Calmette-Guérin vaccine, or microbial component such as β -glucan (Cirovic et al., 2020 [↗](#); Kalafati et al., 2020 [↗](#); Quintin et al., 2012 [↗](#)). Trained immune cells exhibit a rapid and enhanced response to secondary related or unrelated stimulus, relying on chromatin remodeling and metabolic rewiring (Bekkering et al., 2018 [↗](#); Girovic et al., 2020 [↗](#); Su et al., 2021b [↗](#)). In trained cells, histone modification associated with active transcription such as H3-histone-lysine-4 trimethylation (H3K4me3) is enriched in promoter regions of genes encoding inflammatory cytokines, supporting fast gene induction in response to secondary stimulation (Su et al., 2022 [↗](#)). Trained cells typically undergo metabolic rewiring characterized by enhanced glycolysis while the oxidative phosphorylation level shows context-dependent variations (Cheng et al., 2014 [↗](#); Keating et al., 2020 [↗](#)). Intermediate metabolites in tricarboxylic acid (TCA) cycle such as fumarate and succinate function as DNA or histone demethylase inhibitors (Arts et al., 2016 [↗](#)), whereas methionine and SAM in one carbon metabolism act as substrates for methyltransferases (Ampomah et al., 2022 [↗](#); Yu et al., 2019 [↗](#)). Therefore, metabolic processing and metabolites intertwined with epigenetic regulation by directly supplying methylation substrates or regulating methylation enzyme activities. SAM, a ubiquitous methyl donor in biological process, can be converted into S-adenosylhomocysteine (SAH) for the synthesis of cysteine in glutathione metabolism. The ratio of SAM/SAH is critical in support histone H3 trimethylation at lysine 36 (H3K36me3) to enhance IL-1 β transcription (Yu et al., 2019 [↗](#)). However, it has not been clear how SAM level is regulated in the trained immunity process.

Aurora kinase A (AurA), a serine/threonine kinase, plays a critical role in mitosis and is frequently overexpressed in tumor tissues. Inhibition of AurA in tumor cells restricts cell proliferation, migration and induces cell death (Donnella et al., 2018 [↗](#); Jingtai et al., 2023 [↗](#); Tham et al., 2024 [↗](#); Wang-Bishop et al., 2019 [↗](#)). Despite the well-established role of AurA on tumor cells' growth and tumorigenesis, its function in innate immune cells like macrophages or in inflammation is not well understood (L. Ding, Gu, Gao, Xiong, & Zheng, 2015 [↗](#)). A previous study reports that alisertib, a specific inhibitor of AurA, promotes gene expression of demethylase *KDM6B* during THP-1 cell (a human monocyte-like cell line) differentiation (Park, Cho, Oh, Kim, & Seo, 2018 [↗](#)). Whether AurA plays a similar role in regulating trained immunity in macrophages remains unknown. Trained immunity can lead to aberrant inflammatory activity as well as enhanced anti-tumor effect (T. Wang et al., 2023 [↗](#)). Understanding the role of AurA in trained immunity would enable us to further exploit AurA for clinical application in cancer therapy.

In this study, we screened an epigenetic drug library and found that AurA inhibitors notably attenuated the trained immunity induced by β -glucan. Genetic disruption of AurA also showed a similar inhibitory effect on trained immunity. Mechanistically, AurA inhibition reduced the activation of mTOR signaling, thus leading to the nuclear localization of transcription factor FOXO3. Nuclear FOXO3 promoted the expression of GNMT to decrease the intracellular level of SAM, thereby inhibiting H3K4me3 and H3K36me3 enrichment on the promoters of inflammatory genes such as *Il6* and *Tnf*. Moreover, pretreatment with alisertib abolished the anti-tumor effect conferred by β -glucan-induced trained immunity *in vivo*. Thus, we conclude that AurA is an essential kinase required for the epigenetic regulation in macrophages and antitumor activity of trained immunity.

Results

Inhibition of Aurora kinase A suppresses trained immunity in macrophages

Trained immunity is orchestrated by epigenetic reprogramming, while the specific epigenetic modulators involved in regulating trained immunity remain incompletely understood. To uncover the key epigenetic modulators in trained immunity, we performed a small molecule inhibitor screening by using an epigenetic drug library (**Figure 1—figure supplement 1A** [↗](#), Appendix 1—table 1). Briefly, bone marrow-derived macrophages (BMDMs) were trained by β -glucan in the presence of different inhibitors for 24 h, rested for 3 days, and then restimulated by LPS for secondary response. As trained immunity in macrophages strongly induces cytokines production including IL-6 and TNF upon restimulation by pathogen- or damage-associated molecular patterns (PAMPs or DAMPs) (Chakraborty et al., 2023 [↗](#); Netea et al., 2020 [↗](#)), we used IL-6 level from the supernatant as a readout in our screening. The targets of the epigenetic drug library were primarily categorized into several major classes, including Aurora kinase family, histone methyltransferase and demethylase (HMTs and KDMs), acetyltransferase and deacetylase (HDACs and SIRT5), JAK-STAT kinase family, AKT/mTOR/HIF, PARP family, and BRD family (**Figure 1—figure supplement 1B** [↗](#)). Consistent with previous studies, the drug screening results indicated that most inhibitors targeting PI3K/AKT/mTOR and HIF1 α exhibited inhibitory effects on trained immunity (Cheng et al., 2014 [↗](#); de Graaf et al., 2021 [↗](#); Sohrabi et al., 2018 [↗](#)). Furthermore, cyproheptadine, an inhibitor of SETD7, also suppressed trained immunity in our screening system (Keating et al., 2020 [↗](#)). Among total of 305 inhibitors in the compound library, 8 AurA inhibitors showed inhibitory effect on trained immunity-induced IL-6 production in multiple rounds of screening with different concentrations (**Figure 1—figure supplement 1C** [↗](#)). To further confirm whether AurA identified in our screening could modulate trained immunity, we optimized β -glucan training protocol and used three AurA-specific inhibitors, and found that all these AurA inhibitors significantly inhibited IL-6 production in β -glucan-trained mouse macrophages (**Figure 1—figure supplement 1D** [↗](#)). Among these inhibitors, alisertib is an orally active and highly selective inhibitor for AurA and has been applied in preclinical investigation (Bavetsias & Linardopoulos, 2015 [↗](#); Mosse et al., 2019 [↗](#); O'Connor et al., 2019 [↗](#)). We further observed that alisertib obviously downregulated IL-6 and TNF in trained BMDMs in a concentration-dependent manner without affecting cell viability (**Figure 1A–B** [↗](#)). Moreover, alisertib also decreased the transcriptional level of *Il6* and *Tnf* in trained BMDMs (**Figure 1C** [↗](#)). Furthermore, the phosphorylation of AurA was increased by β -glucan but was blocked by alisertib (**Figure 1D** [↗](#)). To further confirm the role of AurA in trained immunity, we knocked down the expression of AurA by small interfering RNAs (siRNAs) in BMDMs (**Figure 1E** [↗](#)). Consistently, knocking down of AurA also inhibited the production levels of IL-6 and TNF in trained BMDMs (**Figure 1F** [↗](#)). Consistent with a recent report showing that tumor cells or tumor cell culture supernatant could function as the second stimulus for trained BMDMs (C. Ding et al., 2023 [↗](#)), we also observed increased TNF and IL-6 production of trained macrophages upon secondary stimulation using MC38 tumor cells culture supernatant, and such effect was also inhibited by AurA knockdown or inhibition (**Figure 1G** [↗](#)). To verify whether AurA regulates trained immunity in different cell models, we trained J774A.1 cells as well as THP-1 cells with β -glucan and also observed a reduction of IL-6 and TNF levels under AurA knockdown or inhibition (**Figure 1—figure supplement 1E** [↗](#)). Moreover, intraperitoneal administration of alisertib together with β -glucan also attenuated trained immunity *in vivo* (**Figures 1H–I** [↗](#)). Collectively, these results suggest that AurA inhibition suppresses β -glucan-induced trained immunity both *in vitro* and *in vivo*.

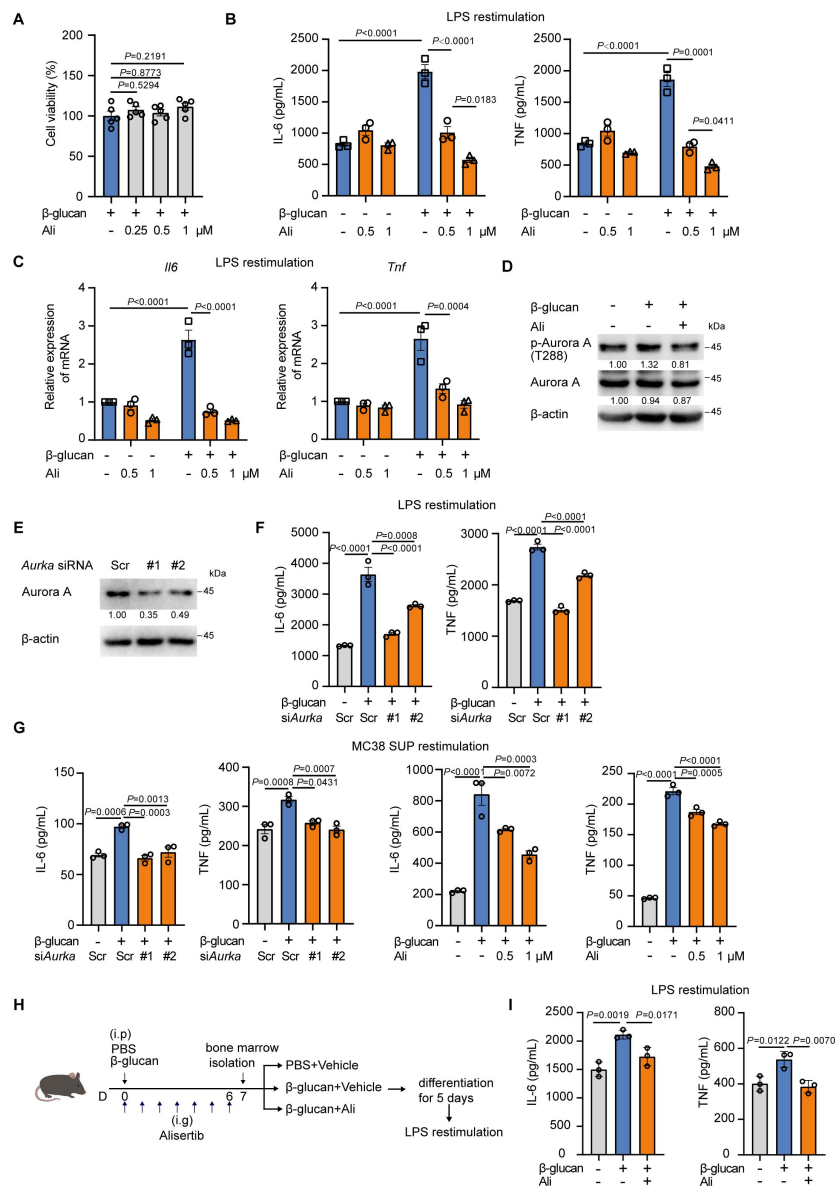


Figure 1.

Inhibition of Aurora kinase A suppresses trained immunity in macrophages.

(A) BMDMs were trained with β -glucan at a dosage of 50 μ g/mL in the presence of different concentration of alisertib for 24 h. The viability of BMDMs was measured by CCK8 assay. **(B)** Supernatant levels of IL-6 (left) and TNF (right) in trained BMDMs with alisertib (0.5 μ M or 1 μ M), followed by restimulation with LPS (100 ng/mL) for 24 h. **(C)** qPCR analysis of relative mRNA expression of *Il6* and *Tnf* in trained BMDMs with alisertib (0.5 μ M or 1 μ M), followed by restimulation with LPS (100 ng/mL) for 6 h. *Actb* served as a reference gene. **(D)** Immunoblotting analysis of AurA phosphorylation after the treatment of β -glucan (50 μ g/mL) with or without alisertib (1 μ M) for 90 min. **(E)** Immunoblotting analysis of AurA in BMDMs transfected with siRNA targeting AurA for 48 h. **(F)** The BMDMs were firstly transfected with siRNAs for 48 hours and then stimulated with β -glucan (50 μ g/mL). Supernatant levels of IL-6 and TNF were detected by ELISA after 3 days rest and restimulation with LPS (100 ng/mL) for 24 h. **(G)** The BMDMs was stimulated with β -glucan (50 μ g/mL) together with AurA knockdown or alisertib, followed by a rest for 3 days and restimulation with MC38 cell culture supernatant for 48 h. **(H)** Graphical outline of *in vivo* training model (3 mice per group). **(I)** Supernatant levels of IL-6 (left) and TNF (right) in trained BMDMs as shown in **H**. Data are presented as the mean $\pm$ SEM (except mean $\pm$ SD for *in vivo* training in **I**). *P* values were derived from one-way ANOVA with Dunnett's multiple-comparison test (A, F, G, I), compared with only β -glucan stimulation group; or two-way ANOVA with Tukey's multiple-comparison test (B, C), compared with every other group. In **D**, similar results were obtained from three independent experiments. Related to **Figure 1—figure supplement 1**, Figure 1—source data 1-2.

Aurora kinase A inhibition remodels chromatin landscape of inflammatory genes

Epigenetic reprogramming towards an open chromatin status is considered the basis for innate immune memory (Arts et al., 2016; Jeljeli et al., 2019; Moorlag et al., 2024; Novakovic et al., 2016). To clarify the status of chromatin landscape under AurA inhibition, we performed an assay for transposase-accessible chromatin with high-throughput sequencing (ATAC-seq) in trained BMDMs. In detail, mice were trained by intraperitoneal injection with β -glucan with or without alisertib, and the bone marrow cells were isolated and differentiated into BMDMs for ATAC-seq analysis (Figure 2—figure supplement 1A). Principal component analysis showed divergent distribution between AurA-treated and -untreated trained BMDMs (Figure 2A). The erased regions by AurA inhibition were mapped into known AurA-regulated cellular processes like “regulation of growth” (Willems et al., 2018), and inflammatory response-related processes like “myeloid leukocyte activation” and “MAPK cascade” (Figure 2B). Analysis of enriched transcriptional binding motifs indicated that the erased peaks by alisertib were transcriptionally regulated by IRF1/2, FOS and STAT1/2, while the written peaks were transcriptionally regulated by PPARG, KLF4, ELF4 and FOXO1/3 (Figure 2C). Recent report shows that PPARG and KLF4 were critical transcription factors in controlling genes of macrophage M2 polarization and FOXO3 regulates the expression of multiple genes involved in pathway such as anti-inflammation and oxidative stress resistance (Allemann, Lee, Beer, & Saeedi Saravi, 2023). Consistent with the reduced production of IL-6 and TNF proteins by alisertib, we observed a decreased chromatin accessibility of proinflammatory gene such as *Il6*, *Tnf*, *Cxcl2* and *Il1a* by alisertib intervention (Figure 2D). In contrast, the peaks of genes encoded M2 marker *mrc1* as well as *chil3* were enhanced (Figure 2E). These results support that AurA regulate epigenetic changes in β -glucan-induced trained immunity and its inhibition restricts the chromatin accessibility for inflammatory genes in macrophages.

Meanwhile, we also performed RNA-seq with the trained BMDMs after LPS restimulation. Analysis of the transcriptome revealed that alisertib inhibited genes associated with pro-inflammatory pathway including “JAK-STAT signaling pathway”, “TNF signaling pathway” as well as “NF-kappa B pathway” in trained BMDMs (Figure 2F). In contrast, the up-regulated genes by alisertib were enriched in anti-inflammation pathway such as “FOXO signaling pathway” (Figure 2G). Additionally, differentially expressed transcription factors were analyzed and mapped into Gene Ontology (GO) with enrichment in “negative regulation of Toll-like receptor signaling pathway”, “negative regulation of NLRP3 inflammasome complex assembly” and “negative regulation of interleukin-6 production” (Figure 2—figure supplement 1B–C). Moreover, multiplex chemokine/cytokines array showed that alisertib decreased the chemokine/cytokines in trained BMDMs such as IL-6, TNF, CXCL2 and IL-1 α (Figure 2—figure supplement 1D). Collectively, these results demonstrate that alisertib restrict chromatin accessibility in genes associated with inflammation activation in trained BMDMs.

Alisertib inhibits glycolysis and remodel TCA cycle

Various studies show that metabolic rewiring such as glycolysis is critical for macrophage memory both in meeting energy demands and epigenetic modification (Bekkering et al., 2018; Bhargavi & Subbian, 2024; Liu et al., 2024). To determine the metabolic pathway that AurA may regulates in trained immunity, we analyzed the glucose metabolism in trained BMDMs. As expected, β -glucan increased glycolysis as indicated by higher extracellular acidification rate (ECAR), which was inhibited by alisertib, but no obvious change in oxidative phosphorylation level was detected (Figures 3A–B, Figure 3—figure supplement 1A). In accordance with enhanced glycolysis, U-¹³C-glucose tracer showed that β -glucan treatment upregulated the glucose incorporation into lactate, and this induction was significantly inhibited by alisertib (Figure 3—figure supplement 1B). Meanwhile, metabolites in TCA cycle, including malate, citrate, α -KG, fumarate and



Figure 2.

Aurora kinase A inhibition remodels chromatin landscape of inflammatory genes.

(A) Principal component analysis (PCA) of gene peaks in ATAC-seq (n=2 mice per group). (B) GO enrichment analysis of erased peaks by alisertib in trained BMDMs. (C) Representative motifs in the erased (n=15,431) and written (n=19,733) peaks respectively. (D and E) Genome browser views of ATAC-seq signal of representative genes inhibited by alisertib including *Cxcl2*, *Il1a*, *Tnf* and *Il6* (D) and representative genes enhanced by alisertib including *Mrc1* and *Chil3* (E). (F and G) KEGG enrichment of differentially expressed genes in trained BMDMs rechallenged with LPS; alisertib downregulated genes (F) and upregulated genes (G) were mapped into KEGG respectively. Related to **Figure 2—figure supplement 1**.

succinate, were all decreased in alisertib-treated trained BMDMs (**Figure 3—figure supplement 1B** [↗](#)). Previous study showed that fumarate was accumulated in β -glucan treated cells, and it induced epigenetic reprogramming to facilitate trained immunity ([Arts et al., 2016](#) [↗](#)). However, alisertib treatment did not affect fumarate accumulation induced by β -glucan (**Figure 3C** [↗](#)), but increased tyrosine level (**Figure 3—figure supplement 1C** [↗](#)). As a recent report shows tyrosine could be interconverted to fumarate ([J. Li et al., 2023](#) [↗](#)), we speculated the decreased fumarate from TCA cycle by alisertib may be compensated from tyrosine metabolism. Taken together, AurA inhibition inhibits glycolysis and remodels glucose metabolism in trained immunity.

Aurora kinase A regulates S-adenosylmethionine level in trained BMDMs

Next, we planned to identify the specific metabolites that support the function of AurA in trained immunity. KEGG analysis by using the differently expressed genes (DEGs) showed that glutathione (GSH) metabolism was significantly enriched (**Figure 3D** [↗](#)). We further found that the intracellular level of GSH in trained BMDMs was notably reduced after AurA inhibition (**Figure 3E** [↗](#)). It has been reported that β -glucan induces a modified cellular redox status as elevation of ROS level and enhanced synthesis of GSH ([Ferreira et al., 2021](#) [↗](#); [Ferreira et al., 2023](#) [↗](#)). Genetic deletion of genes involved in GSH synthesis disrupts the cellular redox balance and dampens trained immunity ([Su et al., 2021a](#) [↗](#)). To investigate whether AurA functions on trained immunity directly through decreasing GSH, we measured the level of serine and SAM, both of which are precursors of GSH (**Figure 3—figure supplement 1D** [↗](#)). By targeted liquid chromatography-tandem mass spectrometry (LC-MS) analysis, we found a significant reduction of SAM level under AurA inhibition, while the serine level remained unchanged (**Figure 3F** [↗](#)). Next, we further detected the downstream products of SAM such as SAH and HCY. Consistently, we observed that alisertib also significantly decreased SAH and HCY in trained BMDMs (**Figure 3G** [↗](#)). The decreased intracellular level of these precursors of GSH indicated that GSH was probably not a metabolite that AurA directly acted on, and its reduction was a result from SAM deficiency. SAM is directly linked to epigenetics as a methyl donor. For example, SAM induction upregulates the total intracellular level of histone lysine 36 trimethylation (H3K36me3) and thus increase H3K36me3 modification at the gene region of *Il1 β* to promote IL-1 β expression ([Rodriguez et al., 2019](#) [↗](#); [Yu et al., 2019](#) [↗](#)). Since AurA inhibition resulted in intracellular SAM reduction, we wondered whether SAM upregulation would rescue trained immunity inhibited by alisertib or by AurA knocking down. GNMT is a key enzyme responsible for the conversion from SAM to SAH, and its deficiency blocked the conversion from SAM to SAH and thus leading to the increased ratio of SAM to SAH ([Hwang et al., 2021](#) [↗](#); [C. H. Li et al., 2015](#) [↗](#); [Yen, Lin, Chen, Chen, & Chen, 2013](#) [↗](#)). Furthermore, we found that AurA inhibition upregulated GNMT protein level in trained BMDMs (**Figure 3H** [↗](#)). To verify whether intracellular SAM level was responsible for AurA-mediated trained immunity, we knocked down GNMT in BMDMs (**Figure 3I** [↗](#)), and found that knockdown of GNMT increased the level of SAM while decreased the level of SAH, leading to increased ratio of SAM/SAH under AurA inhibition (**Figure 3J** [↗](#)). Furthermore, knockdown of GNMT rescued the IL-6 and TNF production in trained immunity under AurA inhibition in response to MC38 cell culture supernatant and LPS (**Figure 3K** [↗](#)). These results suggest that AurA promotes trained immunity by supplying endogenous SAM level which is controlled by GNMT.

Inhibition of Aurora kinase A impairs histone trimethylation at H3K4 and H3K36

SAM is a well-known methyl donor for nearly all cellular methylation events including DNA, RNA and histone methylation ([Keen & Taylor, 2004](#) [↗](#)). Recent study reports that SAM deficiency limits histone methylation by phosphorylation of Rph1, which is a demethylase for H3K36me3 ([Ye et al., 2019](#) [↗](#)). Meanwhile, SAM prefers to induce histone methylation changes in targeted sites like K4 without global changes to DNA and RNA ([Pham et al., 2023](#) [↗](#)). Therefore, we asked whether

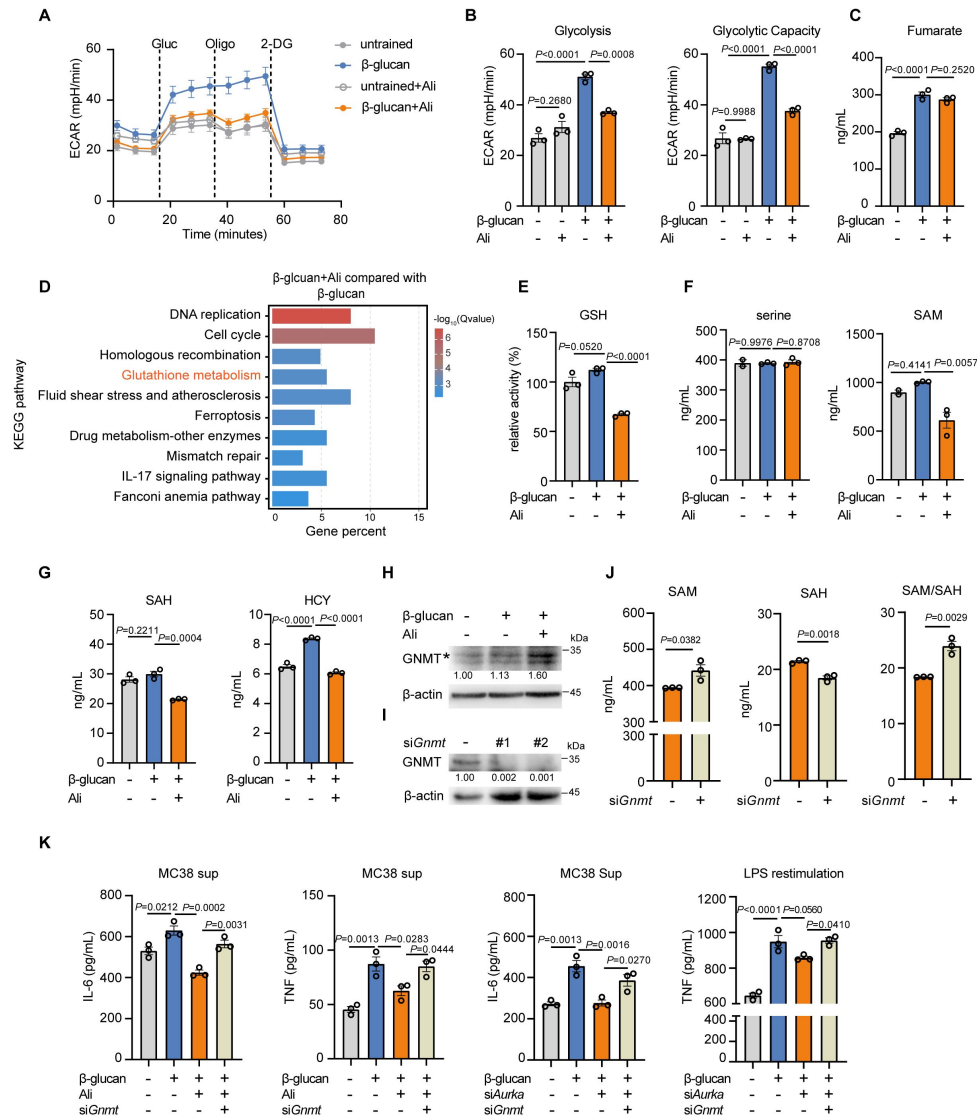


Figure 3.

Aurora kinase A inhibition decreases glycolysis and SAM level. (A and

B) Extracellular acidification rate (ECAR) in BMDMs with different treatments after a glycolysis stress test upon sequential addition of glucose (Gluc, 10 mM), oligomycin (Oligo, 1 μM), and 2-deoxyglucose (2-DG, 50 mM), as indicated (**A**); basal glycolysis and maximal glycolysis (**B**). (**C, F and G**) LC-MS/MS measurements of fumarate (**C**), serine and SAM (**F**), SAH and HCY (**G**) in trained BMDMs treated with vehicle or alisertib for 24 h. (**D**) BMDMs were trained with β-glucan (50 μg/mL) with or without alisertib (1 μM) for 24 h. The BMDMs were collected for RNA extraction and followed by RNA-seq (n=2 mice per group). The TOP 10 enriched pathways identified by KEGG enrichment analysis of differentially expressed genes (Foldchange >1.2, FDR<0.05) by comparing trained BMDMs with or without alisertib. (**E**) Intracellular levels of glutathione in trained BMDMs with or without alisertib for 24 h. The level was normalized to untrained BMDMs. (**H**) Western blot analysis of GNMT in trained BMDMs treated with vehicle or alisertib for 24 h. β-actin was used as a loading control; * showed the position of GNMT protein. (**I**) Western blot showing GNMT protein levels in wild type BMDMs that were transfected with siRNA targeting GNMT. (**J**) LC-MS/MS measurements of SAM and SAH in trained BMDMs treated by alisertib together with or without knockdown of GNMT. The SAM/SAH ratio is calculated by SAH normalization. P values were derived from two-tailed t-test. (**K**) Supernatant levels of IL-6 and TNF in trained BMDMs with AurA inhibition by alisertib or by siRNAs targeting AurA or GNMT. N=3 per group. Data are presented as the mean ± SEM. P values were derived from one-way ANOVA with Tukey's multiple-comparison test (B, K) compared with every other group or with Dunnett's multiple-comparison test (C, E-G) compared with only β-glucan stimulation group. In **H** and **I**, similar results were obtained for three independent experiments. Related to **Figure 3—figure supplement 1**, Figure 3—source data 1-2.

changes in histone methylation occurred when intracellular SAM level was inhibited by AurA inhibition. Among the major active histone methylation markers and repressive markers, we found that trimethylation at H3K4 and H3K36, induced by β -glucan stimulation, were decreased by alisertib. Meanwhile, alisertib did not affect trimethylation at H3K9 and monomethylation at H3K4, accompanied with a modest change but no significant difference in trimethylation at H3K27 (Figure 4A, Figure 4—figure supplement 1 A). Epigenetic modification including H3K4me3 and H3K36me3 promotes chromatin accessibility thus promoting rapid transcription of genes in response to second stimulation in trained immunity. This prompted us to investigate whether alisertib dampened β -glucan-induced epigenetic modification in genes like *Il6* and *Tnf*. Chromatin immunoprecipitation (ChIP) assays demonstrated that both H3K4me3 and H3K36me3 enrichment on *Il6* and *Tnf* promoters induced by β -glucan were notably decreased by alisertib (Figure 4B). Consistently, knockdown of GNMT recovered the expression level of histone methylation and the enrichment level of H3K4me3 as well as H3K36me3 on gene promoter of *Il6* and *Tnf*, which was inhibited by alisertib (Figures 4B–C). Therefore, suppression of trained immunity caused by AurA inhibition is a consequence of decreased trimethylation events in histone by SAM deficiency.

Aurora kinase A regulates GNMT through transcription factor FOXO3

Given the role of GNMT in affecting SAM level, we questioned how AurA inhibition mediated GNMT expression in trained immunity. It was reported that nuclear FOXO3 facilitated GNMT expression to regulate the cell redox response (Hwang et al., 2021). Motif analysis of the ATAC-seq indicated that AurA inhibition promoted FOXO signal (Figure 2C). Thus, we hypothesized that AurA inhibition upregulated GNMT expression via FOXO3. Firstly, we found that knockdown of FOXO3 inhibited GNMT protein level in BMDMs (Figure 5A). Meanwhile, FOXO3 knockdown prevented the enhanced GNMT expression under AurA inhibition and knockdown (Figures 5B–C), indicating the involvement of FOXO3 in AurA function. Moreover, knockdown of FOXO3 also restored the trained immunity as indicated by elevation of IL-6 level, which was inhibited by alisertib (Figure 5D). To further investigate how AurA regulates FOXO3, we detected the phosphorylation level of FOXO3. The result showed that β -glucan stimulation induced the phosphorylation of FOXO3 at Ser315 and AurA inhibition prevented the phosphorylation induction by β -glucan (Figure 5E). Consistently, immunofluorescence detection showed a higher ratio of nuclear enrichment of FOXO3 by AurA inhibition (Figure 5F). Nuclear enrichment of FOXO3 is inhibited by AKT-mTOR activation (Liu et al., 2018; van der Vos et al., 2012). To demonstrate whether the increased FOXO3 nuclear localization by alisertib was associated with AKT-mTOR activation, we checked the activation of AKT-mTOR in trained BMDMs. As expected, both pharmacological and genetic disruption of AurA significantly inhibited β -glucan-induced phosphorylation of AKT-mTOR-S6K-S6 (Figure 5G). Additionally, mTOR activation by its agonist MHY1485 promoted trained immunity under aurora kinase A inhibition (Figure 5H), suggesting that mTOR-FOXO3 signaling pathway acts downstream of AurA-mediated trained immunity. In conclusion, these results show the role of transcription factor FOXO3 in AurA-mediated GNMT expression in trained immunity, which depends on the activation of mTOR pathway.

Alisertib abrogates the anti-tumor effect induced by trained immunity

Trained immunity has been implicated in anti-tumor immunity and is considered as a new branch for cancer immunotherapy development (Bird, 2023; C. Ding et al., 2023; T. Wang et al., 2023). We speculated about whether inhibition of AurA would disrupt the anti-tumor effect induced by β -glucan. Therefore, we conducted a tumor model in mice preinjected with β -glucan (Figure 6A). Training with β -glucan inhibited MC38 tumor growth, but the administration of alisertib abolished the antitumor effect induced by β -glucan (Figure 6B, Figure 6—figure supplement 1 A). Given that trained immunity mediates long-term effects on myeloid cells via

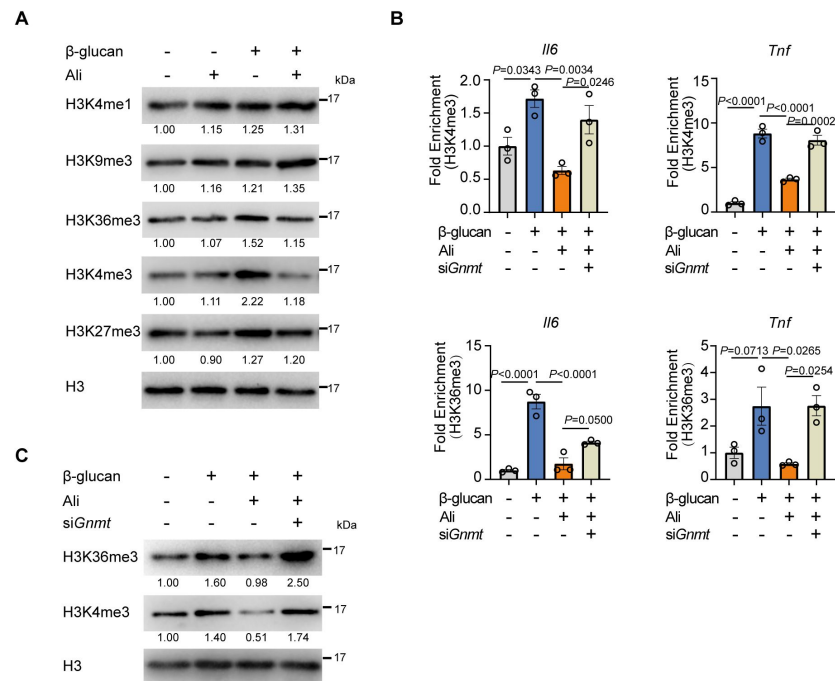


Figure 4.

Inhibition of Aurora kinase A impairs histone trimethylation at H3K4 and H3K36.

(A) Western blot analysis of histone methylation levels in trained BMDM treated with vehicles or alisertib. Histone 3 (H3) was used as a loading control. BMDMs were trained with β-glucan (50 μg/mL) with or without alisertib (1 μM) for 24 h, then BMDMs were washed and cultured in fresh medium for 3 days, followed by protein extraction and Western blot analysis. **(B)** ChIP-qPCR analysis of H3K4me3 and H3K36me3 enrichment in *Il6* and *Tnf* promoter regions in trained BMDMs treated with vehicles or alisertib for 24 h and rest for 3 days. N=3 per group. **(C)** Western blot analysis of total H3K4me3 and H3K36me3 levels upon GNMT knockdown in BMDMs. The BMDMs were transfected with siRNA targeting GNMT for 48 h, followed by β-glucan (50 μg/mL) with or without alisertib (1 μM) treatment for 24 h. Then the BMDMs were washed and cultured in fresh medium for 3 days and the protein was extracted for Western blot analysis. Data are presented as the mean ± SEM. *P* values were derived from one-way ANOVA with Tukey's multiple-comparison test compared with every each other. In **A** and **C**, similar results were obtained for three independent experiments. Related to **Figure 4—figure supplement 1**, Figure 4—source data 1-2.

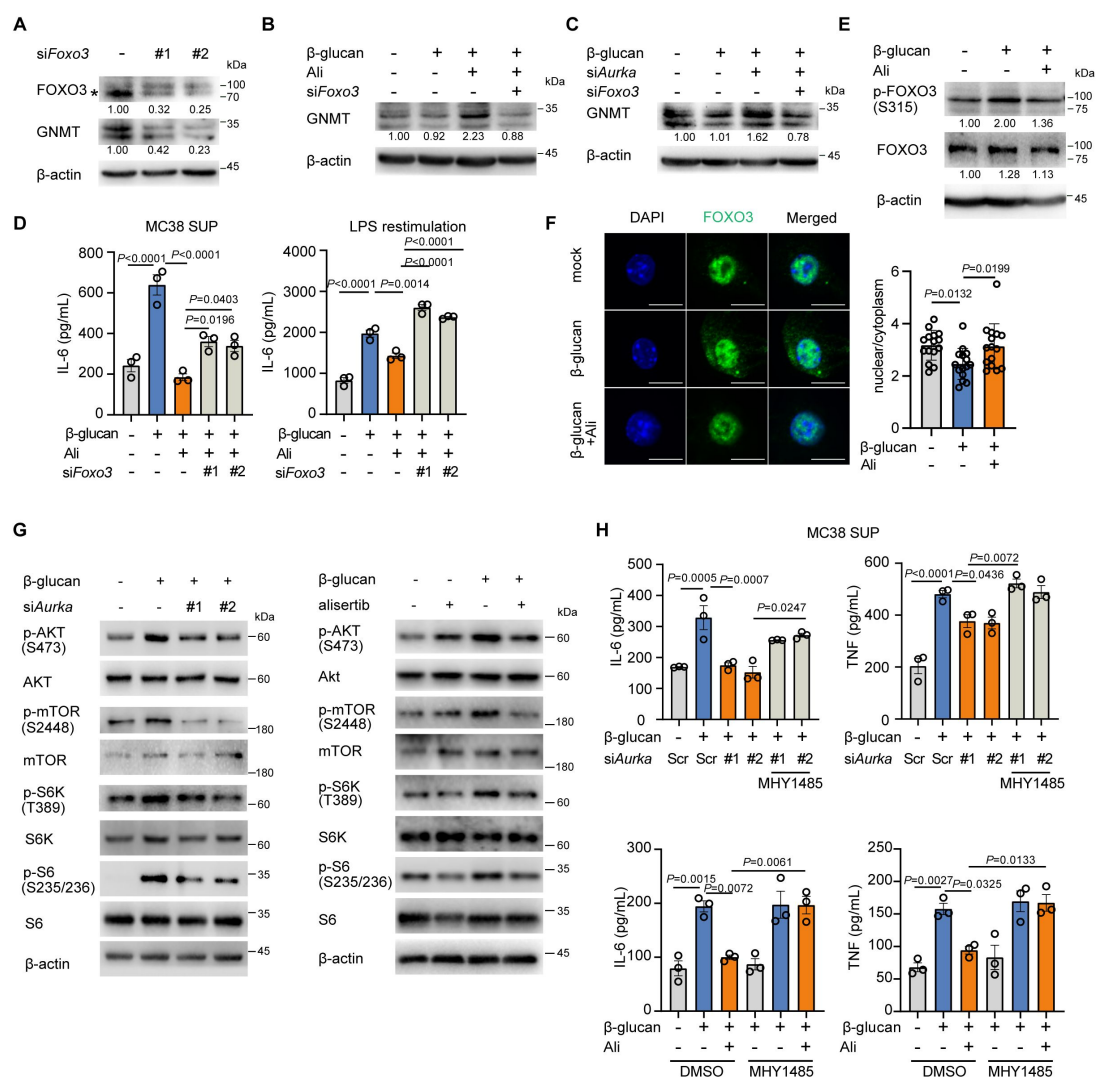


Figure 5.

Aurora kinase A regulates GNMT through transcription factor FOXO3.

(A) Protein level of GNMT in BMDMs with FOXO3 knockdown was detected by Western blot; * showed the position of FOXO3 band. (B and C) Western blot analysis of GNMT downregulation by siFoxo3 in trained BMDMs with AurA inhibition. BMDMs were transfected with siRNA targeting FOXO3 for 48 h, followed by β-glucan (50 μg/mL) and alisertib (1 μM) for 24 h (B); BMDMs were co-transfected with siRNAs targeting FOXO3 and AurA for 48 h, followed by β-glucan (50 μg/mL) stimulation for another 24 h (C). (D) Supernatant levels of IL-6 in BMDMs. The cells were treated with β-glucan (50 μg/mL) and alisertib (1 μM) after transfection of siRNAs targeting FOXO3. (E) Western blot analysis of phosphorylation level of FOXO3 at ser 315 in BMDMs treated with β-glucan (50 μg/mL) with or without alisertib (1 μM) for 12 h. (F) Immunofluorescence staining of FOXO3 in BMDMs after 12 h β-glucan stimulation with or without alisertib. Scale bars: 10 μm (left). The nuclear localization of FOXO3 was compared by calculating the ratio of mean nuclear intensity to cytoplasmic intensity, and the representative data (right) showed the mean intensity of counted macrophages. (G) Western blot analysis of AKT-mTOR-S6 pathway in β-glucan-trained BMDM. BMDMs were transfected with siRNA targeting AurA for 48 h, followed by β-glucan stimulation for 6 h (left); BMDM was trained with β-glucan in the absence or presence of alisertib for 6 h (right). (H) Supernatant levels of IL-6 and TNF in BMDMs. The trained cells were treated with siRNA targeting AurA or alisertib in combination with an mTOR agonist, MHY1485 (2 μM), and restimulated with MC38 culture supernatant for 48 h. Data in D and H are representative of three independent experiments and presented as the mean ± SEM. P values were derived from one-way ANOVA with Tukey's multiple-comparison test compared with each other in D and H, or with Dunnett's multiple-comparison test in F compared with β-glucan only group. In A-C and E-G, similar results were obtained for three independent experiments. Related to Figure 5—source data 1-3.

modulation on bone marrow (BM), we performed bone marrow transplantation (BMT) experiment to directly assess whether alisertib acted on BM to inhibit trained immunity *in vivo*. BM cells were isolated from CD45.1⁺ mice, which were either trained by β -glucan with or without alisertib administration, or left untrained, and transferred to lethally irradiated CD45.2⁺ mice. Four weeks post BMT, CD45.2⁺ mice were inoculated with MC38 tumor cells and tumor growth was monitored (**Figure 6C**, **Figure 6-figure supplement 1B**). Compared to mice received BM cells from untrained mice, the tumor burden was significantly decreased in mice that had received BM cells from trained mice, but not in mice that received BM cells from trained mice with alisertib intervention (**Figure 6D**, **Figure 6-figure supplement 1C**). Therefore, the trained immunity-suppressive properties of alisertib can be transferred by BM transplantation to non-trained recipient mice, which confirmed the direct effect of alisertib on BM-derived cells. Next, we investigated the role of macrophages in the anti-tumor effect induced by β -glucan. We analyzed the population of myeloid cells (CD45⁺CD11b⁺) as well as the population of myeloid derived macrophages (CD45⁺CD11b⁺F4/80⁺) in tumor microenvironment. The population of myeloid cells or macrophage induced by β -glucan or by alisertib intervention showed no changes (**Figures 6E**, **Figure 6-figure supplement 1D**). Studies have shown that GNMT is a tumor suppressor gene and its expression is downregulated in tumor tissue (DebRoy et al., 2013; Simile et al., 2022). Considering that AurA inhibition enhanced GNMT expression in BMDMs in our experiment, we detected GNMT expression in tumor associated macrophages (TAMs). We found that TAMs from alisertib-treated mice exhibited a higher expression of GNMT, compared with β -glucan trained mice (**Figure 6F**). Moreover, compared with alisertib-treated mice, the CD45⁺CD11b⁺F4/80⁺ cells in tumor tissue from β -glucan trained mice showed a higher intracellular phospho-S6, suggesting that AurA inhibition inhibited mTOR activation in TAMs (**Figure 6G**). Pro-inflammatory cytokines induced by β -glucan adjuvant as well as trained immunity can modulate tumor microenvironment and contribute to antitumor immunity (Sui & Berzofsky, 2024; M. Zhang, Kim, & Huang, 2018). As expected, we observed that β -glucan treatment induced a higher level of IL-1 β , IL-6 and IL-12p70 in tumor microenvironment, and the enhanced cytokines production was inhibited by alisertib (**Figure 6H**, **Figure 6-figure supplement 1E**). Moreover, intracellular cytokine staining showed that trained immunity promotes the secretion of IL-6 mainly in infiltrated myeloid cells and macrophages (**Figure 6I**). Taken together, these results demonstrate that AurA inhibition dampens anti-tumor effect of trained immunity via bone marrow cells and reprogrammed the phenotype of tumor-infiltrated myeloid cells including macrophages.

Discussion

Trained immunity defines the memory ability of innate immune cells in response to second challenge. Although the concept of “trained immunity” has been proposed since 2011, the detailed mechanisms that regulate trained immunity are still not completely understood (Netea, Quintin, & van der Meer, 2011). In this study, we identified that AurA was required for trained immunity. AurA deficiency or inhibition decreased IL-6 and TNF production in β -glucan-trained BMDMs upon rechallenge by LPS or supernatant from tumor cells. Moreover, alisertib impaired the anti-tumor effect induced by β -glucan. We also discovered that alisertib induced GNMT expression via mTOR-FOXO3 axis and thus promoted SAM consumption, resulting in decreased SAM/SAH ratio and decreased H3K36me3 and H3K4me3 modification on the promoters of inflammatory genes, highlighting the role of AurA kinase as hub for methionine metabolism and epigenetics.

S-adenosylmethionine, is one of the most ubiquitous metabolites in host. It contains methyl group, adenosyl group and ACP group, all of which can be transferred by corresponding transferases (Lee, Ren, Jeon, & Liu, 2023). It is reported that SAM supplement in mammals is consumed for methylation events on DNA or histone proteins, which affects biological process like tissue differentiation and gene expression (Dai, Ramesh, & Locasale, 2020). In macrophages, DNMT transfers methyl group from SAM to DNA and leads to gene repression such as *Il6* and *Dusp4*

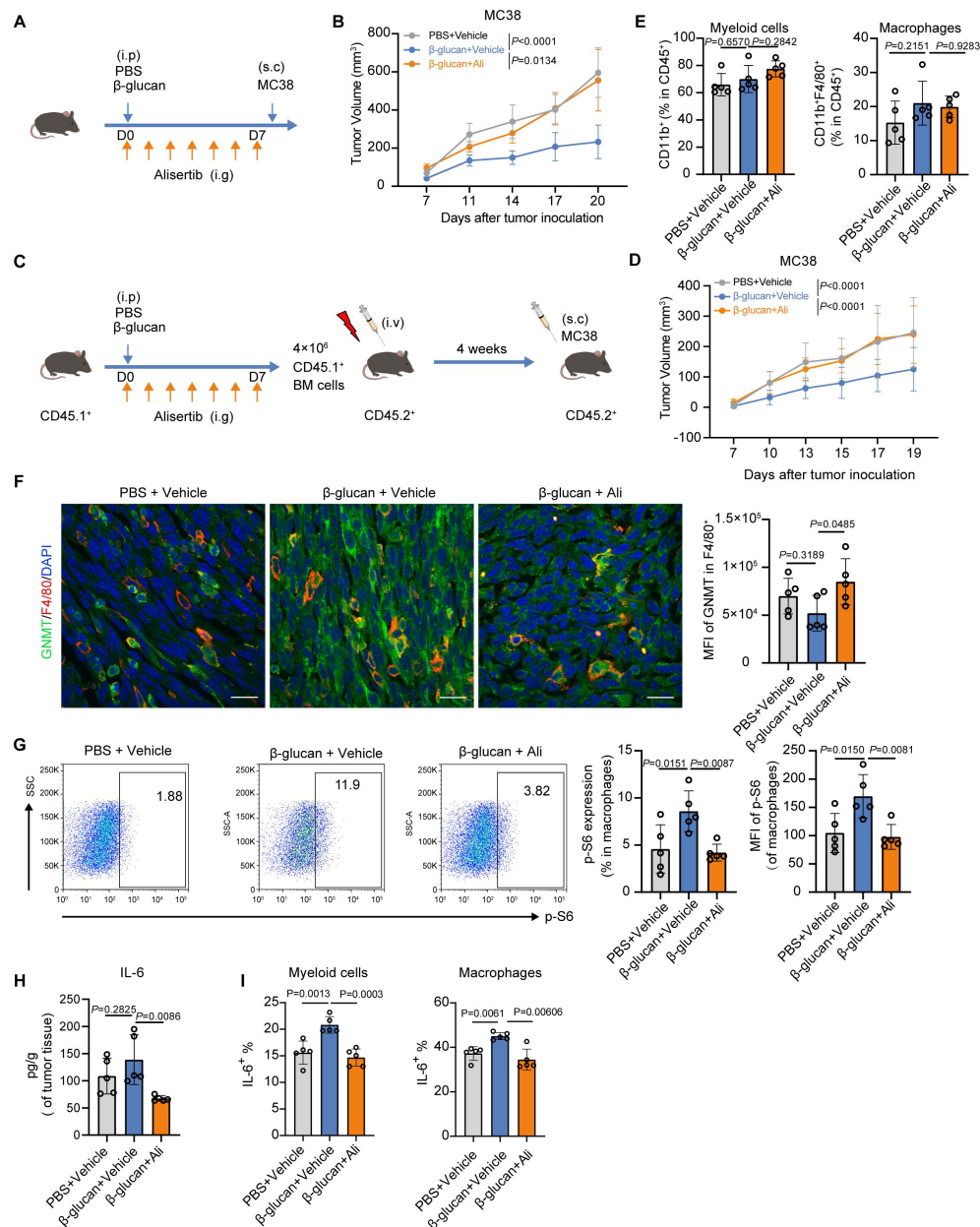


Figure 6.

Alisertib abrogates the anti-tumor effect induced by trained immunity.

(A) Experimental scheme of tumor model. 6~8 weeks old mice were injected with β -glucan together with vehicle or alisertib, followed by subcutaneous inoculation of MC38 cells (1×10^6 cells/mouse, $n=5$ per group). **(B)** The MC38 tumor growth curves of mice in **A**. **(C)** Experimental scheme of bone marrow transplantation model. CD45.1⁺ mice ($n=3$ per group) as donor were treated as indicated, and CD45.2⁺ mice ($n=7$ per group) as recipient received 4×10^6 BM cells from CD45.1⁺ mice at day 8 post irradiation with 9 Gy. After 4 weeks, CD45.2⁺ mice were inoculated with MC38 cells. **(D)** The MC38 tumor growth curves of mice in **C**. **(E)** Flow cytometric analysis of myeloid cells (CD45⁺CD11b⁺) and macrophages (CD45⁺CD11b⁺F4/80⁺) in MC38 tumors in **A**. Gating strategy was shown in Figure 6-figure supplement 1D. **(F)** Co-immunofluorescence staining of DAPI, F4/80 and GNMT in tumor section from **A**; Scale bars: 20 μ m. **(G)** Flow cytometry analysis for intracellular phospho-S6 in macrophages from tumors in **A**. **(H)** Tumor tissue was lysed and the lysates were collected for detection of IL-6 by ELISA. **(I)** FACS analysis of intracellular IL-6 in tumor-infiltrated myeloid cells and macrophages from tumors in **A**. Data are represented as the mean $\pm$ SD. P values were derived from one-way ANOVA (E, F, G, H, I) or two-way ANOVA (B and D) with Dunnett's multiple-comparison test with data from mice treated with β -glucan only as control. Related to **Figure 6—figure supplement 1**, Figure 6—source data 1.

(Ampomah et al., 2022 [↗](#); Ji et al., 2019 [↗](#); Jung, Park, & Ko, 2020 [↗](#)). However, SAM is also reported to play an active role in primary macrophage to increase IL-1 β expression in response to LPS, and its precursor methionine can promote M1 macrophage activation (Dos Santos et al., 2017 [↗](#); Yu et al., 2019 [↗](#)). But whether SAM is involved in trained immunity is not investigated. In our study, AurA inhibition reduced endogenous SAM level in trained BMDMs. It also inhibited the trimethylation at H3K4 and histone H3K36 via upregulating GNMT. GNMT is a glycine methyltransferase, which transfers methyl group from SAM to glycine to lower intracellular SAM level (Luka, Mudd, & Wagner, 2009 [↗](#)). It is well-known that inhibition of AurA would cause abnormal chromatin segregation and chromatin instability. Previous research report that chromatin instability make sensitivity to SAM and polyamine related gene including GNMT, which might also explain the function of AurA inhibition on SAM (Islam et al., 2023 [↗](#)). Additionally, SAM is the precursor for glutathione. It has been reported that methionine and SAM reduce oxidative damage by increasing glutathione (Bandyopadhyay et al., 2022 [↗](#)). However, we found that AurA inhibition didn't increase the level of glutathione, which suggests that alisertib inhibited trained immunity independent of glutathione, but through SAM-mediated epigenetic regulation.

AurA belongs to serine and threonine kinase family. It plays critical role in centrosome and chromosome segregation and cell mitosis. Activation of AurA is associated with tumor progression and drug resistance (Zhao, Zhang, Yang, & Yang, 2023 [↗](#)). During past two decades, there have been developed more than twenty agents as AurA inhibitors (Attwood, Fabbro, Sokolov, Knapp, & Schioth, 2021 [↗](#); Bavetsias & Linardopoulos, 2015 [↗](#)). These agents have been evaluated across metastatic breast cancer, lung cancer, gastro-oesophageal adenocarcinoma and peripheral T-cell lymphoma in clinical trials (Canova et al., 2023 [↗](#); Haddad et al., 2023 [↗](#); Melichar et al., 2015 [↗](#)). Among these inhibitors, alisertib is a highly selective inhibitor for AurA. However, alisertib has limited efficacy against both solid and hematological tumors (Beltran et al., 2019 [↗](#); Mosse et al., 2019 [↗](#); O'Connor et al., 2019 [↗](#)). In recent years, there are researches trying to explain the resistance to AurA-targeted therapy and reveal feedback loops existed in tumor cells, which contributes to drug resistance. For example, AurA inhibition by alisertib upregulates PD-L1 expression in tumor cells and allows immune escape (X. Wang et al., 2023 [↗](#)). AurA inhibition also enhances fatty acid oxidation to overcome the glycolysis attenuation in tumor cells (Nguyen et al., 2021 [↗](#)). However, these studies mainly focus on investigating the role of AurA in tumor cells while its role in immune cells is rarely understood. In this study, we identified AurA as a key regulator in promoting trained immunity through epigenetic drug screening. And AurA inhibition significantly decreased the trained immunity induced by β -glucan *in vitro* and *in vivo*. AurA is described to be expressed and activated in stem-cell like cells, bone marrow and epithelium (Qi et al., 2016 [↗](#); Rio-Vilarino et al., 2024 [↗](#); Zhou et al., 2018 [↗](#)). Clinical data show that alisertib by oral administration rapidly distributes in bone marrow, making bone marrow a site susceptible to side effect (Oh, Power, Zhang, Daniels, & Elmquist, 2022 [↗](#)). Due to the inhibition effect of alisertib on trained immunity in bone marrow, this may be the potential explanation for the poor therapeutic efficacy or drug resistance of alisertib on patients with cancer. In our primary drug screening, all compounds were examined at a fixed concentration of 5 μ M. Despite this limitation, our screening results recapitulated several trained immunity inhibitors previously reported in literature, thereby reinforcing the validity of our screening and highlight some potential molecular targets worthy of further investigation. Overall, our findings provide a potential guidance for AurA inhibitor application in clinical cancer therapy, and the clues for design of new inhibitors such as avoiding targets on bone marrow.

In conclusion, our findings demonstrate that AurA supports trained immunity by maintain SAM level. As a result of SAM deficiency under AurA inhibition, H3K4m3 and H3K36me3 level are reduced and trained immunity is inhibited. Our finding reveals a novel AurA-SAM metabolic axis as a new mechanism for trained immunity. Furthermore, our findings also identify a potential guidance for the clinical application of AurA, and new clues for the design of next-generation AurA inhibitors in the future.

Materials and methods

Mice

C57BL/6J mice (male, 6~8 weeks old) were purchased from Vital River Laboratory Animal Technology Co., Ltd. (Beijing, China). All mice were kept in specific pathogen-free (SPF) conditions. For all animal studies, mice were randomly assigned to experimental groups, and the sample size was determined according to previous similar study in tumor models (Arifin & Zahiruddin, 2017 [\[4\]](#)). All Animal experiments were performed according to protocols approved by the Institutional Animal Care and Use Committee of Sun Yat-sen University. Mice were injected intraperitoneally with 2 mg β -glucan (Cat#G6513, Sigma-Aldrich, USA) suspended in 200 μ L PBS. For drug administration, alisertib (Cat#A4110, ApexBio, shanghai, dissolved in 10% 2-hydroxypropyl- β -cyclodextrin and 1% sodium bicarbonate, oral gavage) was delivered once daily at 30 mg/kg for 7 days before MC38 inoculation. MC38 (1×10^6 cells/mouse) were subcutaneously inoculated into the mice. The tumor volumes were measured every 2 or 3 days, and were calculated as: $1/2 \times \text{length} \times \text{width}^2$.

Transplantation and repopulation assay

CD45.1⁺ B6/SJL male mice (5~6 weeks old) as donor mice were purchased from Shanghai Model Organisms Center Inc. The mice were injected with of β -glucan (2 mg per mouse) or in combination with daily administration of alisertib (30 mg/kg/d) or vehicle for 7 days. On day 8, BM cells were isolated and 4×10^6 cells were intravenously injected into lethally irradiated CD45.2⁺ C57BL/6J male mice (6~8 weeks old), which received 2 equal doses of 4.5 Gy at least 4 hours apart with X-ray source (RS2000 Plus-225, RadSource Technologies Inc). CD45.2⁺ Mice were kept under antibiotic treatment (0.1‰ levofloxacin,) for one week before and four weeks after irradiation and reconstitution. Four weeks after transplantation, CD45.2⁺ mice received a subcutaneous injection of MC38 cells (1×10^6 cells/mouse). The tumor volumes were measured and recorded every 2 or 3 days.

Cell Culture

THP-1 cells were cultured in 1640 medium with 10% FBS, 1% Pen/Strep and 1% Glutamax. J774A.1 cells were cultured in DMEM medium with 10% FBS, 1% Pen/Strep. For primary macrophages, bone marrows were isolated from C57BL/6J mice (6~8 weeks) and cultured in 1640 medium with 10% FBS, 1% Pen/Strep, 10% supernatant from L929 cells. For transfection with siRNAs (synthesized by RuiBiotech, Guangzhou) using RNAiMAX (Invitrogen, Cat#13778150) according to the manufacturer's instruction, bone marrows cultured for 4 days were digested with trypsin and seeded into cells plates followed by transfection on day 5. Cells was maintained in 95% humidified air and 5% CO₂ at 37°C. All cells were authenticated tested for mycoplasma contamination.

The siRNAs sequences were as follows:

siRNA for mouse AurA#1: 5'-CGAGCAGAGAACAGCUACUUATT-3'

siRNA for mouse AurA#2: 5'-GCACCCUUGGAUCAAAGCUAATT-3'

siRNA for mouse GNMT#1: 5'-GGACAAAGAUGUGCUUUCATT-3'

siRNA for mouse GNMT#2: 5'-CGUCAGUACUGACAGUCAATT-3'

siRNA for mouse FOXO3#1: 5'-CGGCAACCAGACACUCCAATT-3'

siRNA for mouse FOXO3#2: 5'-CUGUAUUCAGCUAGUGCAATT-3'

Scramble siRNA: 5'-UUCUCCGAACGUGUCACGUTT-3'

Drug screening

Drugs screening was performed using a drug library from TargetMol (Shanghai, China). A detailed drugs list was provided in Appendix 1—table 1. For the primary screening, 1×10^5 BMDMs were seeded into 96-wells plate. Following overnight incubation, the BMDMs were trained with β -glucan (100 μ g/mL, Sigma-Aldrich, Cat#G6513) in the presence of drugs at a concentration of 5 μ M for 24 h while the secondary screening for parts of the drugs was performed at two concentrations of 0.2 and 1 μ M. Control cells received vehicle control DMSO. After 24 h training, cells were washed with 1 $\times$ PBS and further cultured in fresh complete medium (containing 10% L929 supernatant) for 3 days, followed by LPS stimulation (1 μ g/mL). The culture supernatant was subjected to ELISA to measure IL-6. The relative amount of IL-6 was calculated as the fold change of the drugs-treated trained cells compared with the glucan only-treated cells. The drugs that showed suppressing effects with fold changes of 0.8-fold or lower, were considered to be inhibitors for trained immunity.

Cell viability assay

The effect of alisertib on cell viability was examined by CCK8 assay (Cat#40203ES60, Yeasen). In brief, BMDMs were seeded in to 96-wells plate and trained with β -glucan (50 μ g/mL) in the presence of alisertib at a different concentration. After 24 h treatment, CCK8 reagent was added to the wells for 4 h, the cell viability was measured at 450 nm.

Western blot analysis

To evaluate intracellular protein expression, total cellular protein was extracted with lysis buffer containing 50 mM Tris-HCl (pH 7.4), 150 mM NaCl, 1% NP-40, 0.1% SDS and 0.5% Na-deoxycholate supplemented with Protease Inhibitor Cocktail (Selleck, Cat#B14001) and Phosphatase inhibitors (Roche). Proteins were resolved by SDS-PAGE and subjected to Western blot as described elsewhere. The primary antibodies were incubated at 4°C overnight and HRP-conjugated secondary antibodies was incubated for 1 h at RT. Enhanced chemiluminescence was used to detect the specific blot bands. The specific bands were quantified using Image Lab (v6.0) and the relative expression was normalized to the internal control.

The following primary antibodies were used: anti- β -actin (1:10000, Cat#A2228, Sigma-Aldrich), anti-Aurora A (1:1000, Cat#ab108353, Abcam), anti-Phospho Aurora A (Thr288) (1:500, Cat#44-1210G, Invitrogen), anti-GNMT (1:500, Cat# PA5-100018, Invitrogen), anti-H3K4me1 (1:3000, Cat#ab8895, Abcam), anti-H3K9me3 (1:3000, Cat#ab8898, Abcam), anti-H3K36me3 (1:3000, Cat#ab282572, Abcam), anti-H3K4me3 (1:3000, Cat# ab213224, Abcam), anti-H3K27me3 (1:3000, Cat#9733, Cell Signaling Technology), anti-H3 (1:5000, Cat#14269, Cell Signaling Technology), anti-FOXO3 (1:500, Cat#ab23683, Abcam), anti-Phospho FOXO3 (Ser315) (1:500, Cat# 28755-1-AP, Proteintech), anti-AKT (1:1000, Cat# 2920, Cell Signaling Technology), anti-Phospho-AKT (Ser473) (1:1000, Cat# 4060, Cell Signaling Technology), anti-mTOR (1:1000, Cat# 2972, Cell Signaling Technology), anti-Phospho-mTOR (Ser2448) (1:1000, Cat# 5536, Cell Signaling Technology), anti-p70 S6 Kinase (1:1000, Cat# 9202, Cell Signaling Technology), anti-Phospho-p70 S6 Kinase (Thr389) (1:1000, Cat# 9205, Cell Signaling Technology), anti-S6 (1:1000, Cat# 2217, Cell Signaling Technology), and anti-Phospho-S6 (Ser235/236) (1:1000, Cat# 4858, Cell Signaling Technology). Goat anti-rabbit HRP-linked antibody (1:3000, 7074, Cell Signaling Technology) or goat anti-mouse HRP-linked antibody (1:3000, 7076, Cell Signaling Technology) served as the secondary antibody.

Bulk RNA sequencing

Total RNA was extracted by TRIzol™ reagent and purified. Library construction and sequencing were performed by Annoroad Gene Technology (Guangzhou). After removal of rRNA and filtering of clean reads, the paired-end reads were mapped to the GRCm38/mm10 genome using HISAT2 2.2.1 (Florea, Song, & Salzberg, 2013 [\[1\]](#)). FPKM was obtained for each gene in the RNA-seq data using HTSeq 2.0.2. RNA differential expression analysis was performed by DESeq2 software (Love, Huber, & Anders, 2014 [\[2\]](#)). The genes with FDR (false discovery rate) below 0.05 and absolute fold change (FC) not less than 1.2 were considered as differentially expressed genes (DEGs). All DEGs were mapped to GO terms in the Gene Ontology database for GO enrichment analysis and were mapped to KEGG for pathway enrichment analysis using ClusterProfiler 4.6.6.

ATAC-seq analysis

1×10^5 cells were prepared for ATAC-seq library construction with Novoprotein Chromatin Profile Kit for Illumina as the manufacturer's instructions. The solubilized DNA fragments were amplified with Novoprotein NovoNGS Index kit for Illumina. The reaction was monitored with qPCR to prevent GC bias and oversaturation. Finally, the library was purified by Novoprotein DNA Clean Beads and sent to Annoroad Gene Technology (Guangzhou) to sequence with Illumina Novaseq 6000 (150 bp, paired-end). ATAC-seq data were analyzed following guidelines of Harvard FAS Informatics Group. Optimally, reads were aligned to the reference mouse genome (GRCm38/mm10) using HISAT2 with X, no-spliced-alignment, and no-temp-splice site parameters according to the manual. Quality control was performed with FastQC (Version 0.12.1) in Linux and with ATACseqQC (Version 1.16.0) in R (Version 4.1.0) (Wilbanks & Facciotti, 2010 [\[3\]](#)). The peaks were called using MACS3 software with default settings tailored for the genome (Y. Zhang et al., 2008 [\[4\]](#)). For all remaining samples, peaks with low counts (consensus peaks with a median across treatments of counts-per-million less than 1) were filtered out from further analysis submitted to Diffbind~3.8.4 to perform further analysis (Ross-Innes et al., 2012 [\[5\]](#)). The read distribution of PCA plot was generated by dba.plotPCA with the normalized count matrix. And the differential analysis were performed by dba.analyze function with DBA_EDGER. The differentially accessible regions (DARs) were defined as log fold changes >1 or <-1 , meanwhile the $p < 0.05$. In order to determine whether differentially accessible chromatin peaks localized near genes with shared functional biological pathways, we performed peak annotation by Genomic Regions Enrichment of Annotations Tool (GREAT) (Heinz et al., 2010 [\[6\]](#)). We used 500kb as the maximum absolute distance to the nearest transcriptional start site and q-values less than 0.01 as statistically significant. TF Motif Enrichment Analysis was performed using HOMER's findmotifsGenome. Integrative Genomics Viewer (IGV) track plots of chromatin accessibility were generated with the bigwig files.

ECAR/OCR measurement and Glutathione detection

Extracellular acidification rate (ECAR) and oxygen consumption rate (OCR) were determined by Seahorse Flux Analyzer XF96 (Agilent) according to the manufacturer's instructions. In brief, BMDMs were plated at a density of 5×10^4 cells per well into XF96 cell culture microplates (Agilent) and cultured overnight followed by β -glucan (50 $\mu\text{g/mL}$) treatment with or without alisertib (1 μM) for 24 h. The stimulated cells were then washed with Seahorse XF RPMI Media (pH 7.4) supplemented with 2 mM glutamine and incubated for 1 h at 37°C without CO₂. The cell culture plate was sequentially injected of glucose (10 mM), oligomycin (1 μM) and 2-DG (50 mM) for measurement of ECAR. For OCR measurement, the plate was injected of oligomycin (1 μM), FCCP (1 mM), and rotenone/antimycin A (0.5 μM). Both analyses of ECAR and OCR were performed using Agilent Seahorse XF96 Analyzer. For intracellular GSH detection, a total of 8×10^6 BMDMs were seeded into 10 cm dishes. Cells were treated by β -glucan (50 $\mu\text{g/mL}$) with or without alisertib (1 μM) for 24 h. Following that, cells were collected by trypsinization and washed once with 1×PBS. Cell pellet was resuspended in 1 mL of PB buffer before sonication. The lysate was centrifuged at

14,000 rpm at 4°C for 10 min, and supernatants was collected for measurement of glutathione according to the manufacturer's instructions (Cat#DIGT-250, BioAssay Systems). The relative GSH level was calculated by normalization to mock group.

U13-glucose tracing

Labelled compounds U-¹³C-glucose were added to customized RPMI medium lacking glucose. And BMDMs were trained with β-glucan (50 μg/mL) plus alisertib (1 μM) for 20 h in this customized medium with labelled U-¹³C-glucose. Following that, cells were washed with 1× PBS twice by aspirating the medium and immediately adding -80°C methanol and pre-cold water containing norvaline. After 20 min of incubation on dry ice, the resulting mixture was scraped, collected into a centrifuge tube, mixed with pre-cold chloroform and centrifuged at 14,000×g for 5 min at 4°C. The supernatant was evaporated to dryness in a vacuum concentrator. The dry residue was completely dissolved in 20 μL of 2% (w/v) methoxyamine hydrochloride in pyridine and incubated for 60 min at 37°C. N-tert-Butyldimethylsilyl-N-methyltrifluoroacetamide with 1% tert-butyldimethylchlorosilane was used for the preparation of N-tert-butyldimethylsilyl ethanolamines (TBDMS) and sample was incubated with TBDMS for 30 min at 45°C for derivatization. After centrifugation for 3 min at 12000 rpm, the derived samples were detected and analyzed using a Thermo1310 coupled to an IQ QD MS system with DB-35 (30 m×0.25 mm×0.25 μm). The ion source is an ionization energy of 70 eV and temperature of 300°C. The detection was in a mode of full scan, ranging between 100-650 m/z. The inlet temperature was 270°C, and helium was used as the carrier gas, in a flow rate of 1.2 ml/min. Polar metabolite ramp-up procedure: The initial temperature of the column heater was maintained at 100°C for 2 min, and the temperature was raised to 255°C at a rate of 3.5°C/min and to 320°C at a rate of 15°C/min, with a total run time of approximately 50 min.

Targeted metabolomics analysis

To measure SAM, SAH, serine, fumarate and HCY, liquid chromatography (LC)–tandem mass spectrometry (MS/MS) analysis was performed. Briefly, BMDMs were trained with β-glucan (50 μg/mL) and inhibited by alisertib for 24 h. For transfection with siRNAs targeting GNMT, BMDMs was transfected with siRNA for 48 h and was followed by β-glucan and alisertib treatment for another 24 h. Metabolites were extracted by using cold 80% methanol (HPLC Grade, Sigma-Aldrich) from 3 million cells with vortex, followed by centrifugation for 15 min at 15000×g at 4°C to collect supernatant. The supernatant was then evaporated to dryness in a vacuum concentrator. Analysis of targeted metabolites was conducted on a A6495 triple quadrupole system interfaced with a 1290 UHPLC system (Agilent Technologies). The resulting MS/MS data were processed using Agilent Quantitative analysis software.

Chromatin immunoprecipitation (ChIP) assay

The SimpleChIP® Enzymatic Chromatin IP Kit (Cat#9003, Cell Signaling Technology, USA) was used to perform ChIP according to the manufacturer's instructions. Samples were subjected to immunoprecipitation using either Rabbit anti-H3K4me3 antibody, anti-H3K36me3 antibody or a control IgG antibody (Cell Signaling Technology). Fragmented DNAs were purified using spin columns (Axygen) and was used as the templates for qPCR using indicated primer sets spanning the *Tnf* and *Il6* promoters.

The following primers for Chip-PCR were used:

Chip *Il6* forward: 5'-TCGATGCTAAACGACGTCACA-3'

Chip *Il6* reverse: 5'-CGTCTTTTCAGTCACTATTAGGAGTC-3'

Chip *Tnf* forward: 5'-TGGCTAGACATCCACAGGGA-3'

Chip *Tnf* reverse: 5'-AAGTTTCTCCCCAACGCAA-3'

Real-time quantitative PCR

BMDMs were stimulated as indicated and the total RNA was extracted using an EZ-press RNA Purification Kit (EZBioscience) according to the instrument. Then total RNA was reverse-transcribed into cDNA using HiScript III RT SuperMix (Cat#R323, Vazyme), followed by real-time quantitative polymerase chain reaction (qPCR) using 2 × PolarSignal SYBR Green mix Taq (Cat#MKG900, MIKX) for analysis of mRNA expression. All data were normalized to *Actb*. Primer sequences were as follows:

Mouse *Actb* forward: 5'-AGAGGGAAATCGTGCCTGAC-3'

Mouse *Actb* reverse: 5'-CAATAGTGATGACCTGGCCGT-3'

Mouse *Il6* forward: 5'-TAGTCCTTCCTACCCCAATTTC-3'

Mouse *Il6* reverse: 5'-TTGGTCCTTAGCCACTCCTTC-3'

Mouse *Tnf* forward: 5'-CCCTCACACTCAGATCATCTTCT-3'

Mouse *Tnf* reverse: 5'-GCTACGACGTGGGCTACAG-3'

Immunocytochemistry and immunohistochemistry staining

For intracellular staining of FOXO3 (1:200, Cat#ab23683, Abcam), BMDMs was grown on glass bottom dishes (Nest, Wuxi) overnight. After indicated treatment, BMDMs were washed twice with 1×PBS, fixed in 4% paraformaldehyde in RT for 10 min, permeabilized with 0.25% Triton X-100 for 30 min at RT. Cells were incubated with blocking buffer (1% BSA, 0.1% Tween 20 in TBS) for 30 min at RT. Primary antibody were incubated at 4°C overnight, and the appropriate fluorescent secondary antibody (1:500, Cat# A-11008, Thermo Fisher Scientific) for 1 h at RT. The dishes were then washed three times with 0.1% TBST with DAPI staining during the first wash. For tumor tissue staining, fresh tumor tissue was fixed in 4% paraformaldehyde in RT for 24~48 h and cut into ~8 μm section after embedded in OCT. The slice was blocked with normal goat serum with 1% Triton X-100 and incubated with primary antibodies including anti-GNMT (1:200, Cat# PA5-100018, Invitrogen), and anti-F4/80 (1:200, Cat#ab90247, Abcam) antibody overnight at 4°C, followed by incubation with Alexa Fluor 488 anti-rabbit (1:200, Cat# A-11008, Thermo Fisher Scientific) and Alexa Fluor 555 anti-rat (1:200, Cat# 4417, Cell Signaling Technology) for 1 h at RT. Nuclei were counterstained with DAPI (1 mg/mL) and mounted with Prolong GLASS antifade mountant (P10144, Thermo Fisher Scientific). Images were obtained using a NIKON confocal microscope. The nuclear localization of FOXO3 in trained BMDMs *in vitro* was compared by calculated the ratio of mean nuclear intensity to cytoplasmic intensity (Kelley & Paschal, 2019 [\[3\]](#)). For analysis the GNMT expression in tumor associated macrophages, we randomly counted 150~200 DAPI⁺F4/80⁺ cells and draw ROI to report intensity. The representative data in our result showed the mean intensity of counted macrophages.

FACS

Fresh tumor tissue was harvested at the end of time point and cut into small pieces. A single-cell suspension was obtained by crushing the tumor tissue through a 100 μm cell strainer in Phosflow lyse/fix buffer (Cat#558049, BD Biosciences) immediately. Subsequently, the suspended cells were washed twice with BD perm/wash buffer (Cat#554723, BD Biosciences). Cells were then stained with the desired antibodies for 30 min at RT in dark. For phospho-S6 staining, cells were incubated with secondary antibody. Cells were analyzed using a BD LSRFortessa flow cytometer. The following primary antibodies were used: anti-CD45 BV421 (1:200, Cat#563890, BD Biosciences),

anti-CD45.1 FITC (1:200, BioLegend, Cat#110705), anti-CD45.2 BV510 (1:200, BioLegend, Cat#109837), anti-CD11b PerCP Cy5.5 (1:200, Cat# 45-0112-82, Thermo Fisher Scientific), anti-F4/80 FITC (1:200, Cat# 11-4801-82, Thermo Fisher Scientific), anti-Phospho-S6 (Ser235/236) (1:200, Cat# 4858, Cell Signaling Technology), and Fixable Viability Dye (1:1000, BioLegend, Cat#423105). Anti-rabbit IgG (Alexa Fluor 555 Conjugate) (1:200, Cat# 4413, Cell Signaling Technology) served as the secondary antibody.

Statistical analysis

The GraphPad Prism version 8.0 was used for statistical analysis. Data are presented as mean $\pm$ SEM (Standard Error of Mean) of indicated biological replicates for *in vitro* cellular experiments, while data are presented as mean $\pm$ SD (Standard Deviation) of indicated biological replicates for *in vivo* experiment. For statistical significance analysis, two-tailed t-test was used for comparing two mean values; one-way ANOVA with Dunnett's multiple-comparison test was applied for comparing three or more mean values with β -glucan group as control; one-way ANOVA or two-way ANOVA with Tukey's multiple-comparison test was applied for comparing multiple mean values under different conditions. The data distribution was assumed to be normal, although this assumption was not formally tested.

Figure supplements

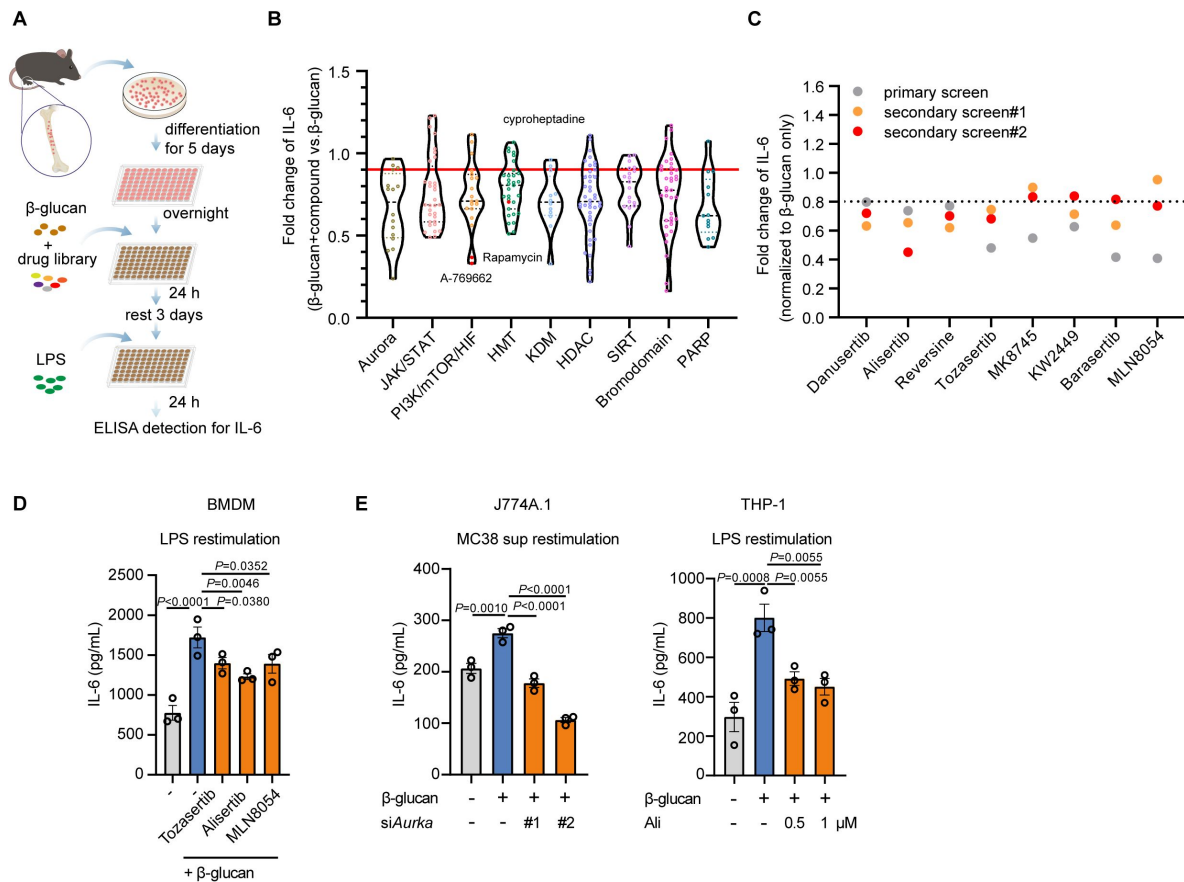


Figure 1—figure supplement 1.

Targeting aurora A inhibits β -glucan-induced trained immunity.

(A) Schematic of the assay protocol for the drug screening in BMDMs. **(B)** Fold change of IL-6 production showing the outcomes of the entire drug screening, including inhibitors targeting Aurora kinase, JAK/STAT pathway, PI3K/AKT/mTOR pathway, histone methyltransferases (HMT), lysine-specific demethylases (KDM), histone deacetylases (HDAC), Sirtuin (SIRT), Bromodomain and PARP families. **(C)** Fold change of IL-6 production inhibited by AurA inhibitors compared to that of β -glucan only group. **(D)** BMDMs were trained with β -glucan (50 μ g/mL) in the presence of Tozasertib (1 μ M), alisertib (1 μ M) or MLN8054 (1 μ M) for 24 h, followed by LPS (100 ng/mL) stimulation for 6 h after a rest for 3 days. **(E)** Supernatant levels of IL-6 in trained J774A.1 cells and THP-1 cells. J774A.1 cells were transfected with AurA-specific siRNAs followed by β -glucan (50 μ g/mL) stimulation. After rest for 3 days, trained J774A.1 cells were counted and seeded into cell culture plate with MC38 supernatant rechallenge for 48 h. THP-1 cells were trained with β -glucan (50 μ g/mL) for 24 h, and were centrifuged and washed once to remove medium. The THP-1 cells were then cultured with fresh medium containing 10 ng/mL PMA for 48 h and rest 1 day, followed by LPS rechallenge (100 ng/mL). Data in **D** and **E** are representative of three independent experiments and presented as the mean $\pm$ SEM. *P* values were derived from one-way ANOVA with Dunnett's multiple-comparison test, compared with β -glucan stimulation only group. Related to [Figure 1](#).

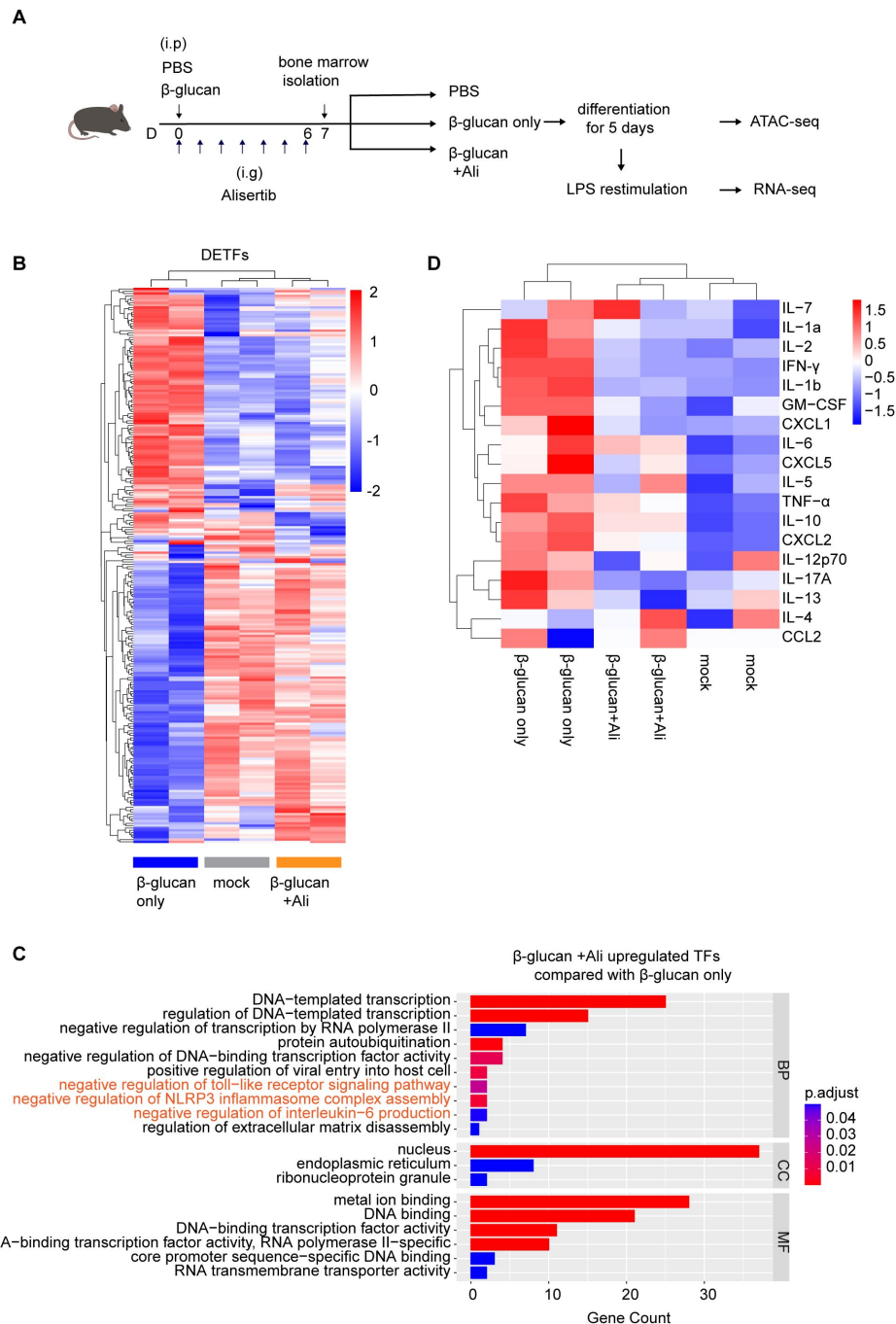


Figure 2—figure supplement 1.

Aurora kinase A inhibition suppresses the expression of transcription factors involved in inflammation activation.

(A) *In vivo* training model in C57BL/6J mice with intraperitoneal injection of β -glucan (2 mg per mice) and daily administration of alisertib (30 mg/kg/d) for 7 days ($n=2$ mice per group). **(B)** Heatmap from RNA-seq analysis showing the differentially expressed transcription factors (DETFs) from mice treated as described in **A**. **(C)** GO enrichment analysis of differentially expressed transcription factors (DETFs). **(D)** Multiplex immunoassay measuring 18 cytokines/chemokines in supernatant from trained BMDMs as described in **A**, which were rechallenged with LPS (100 ng/mL) for 6 h. Related to **Figure 2**.

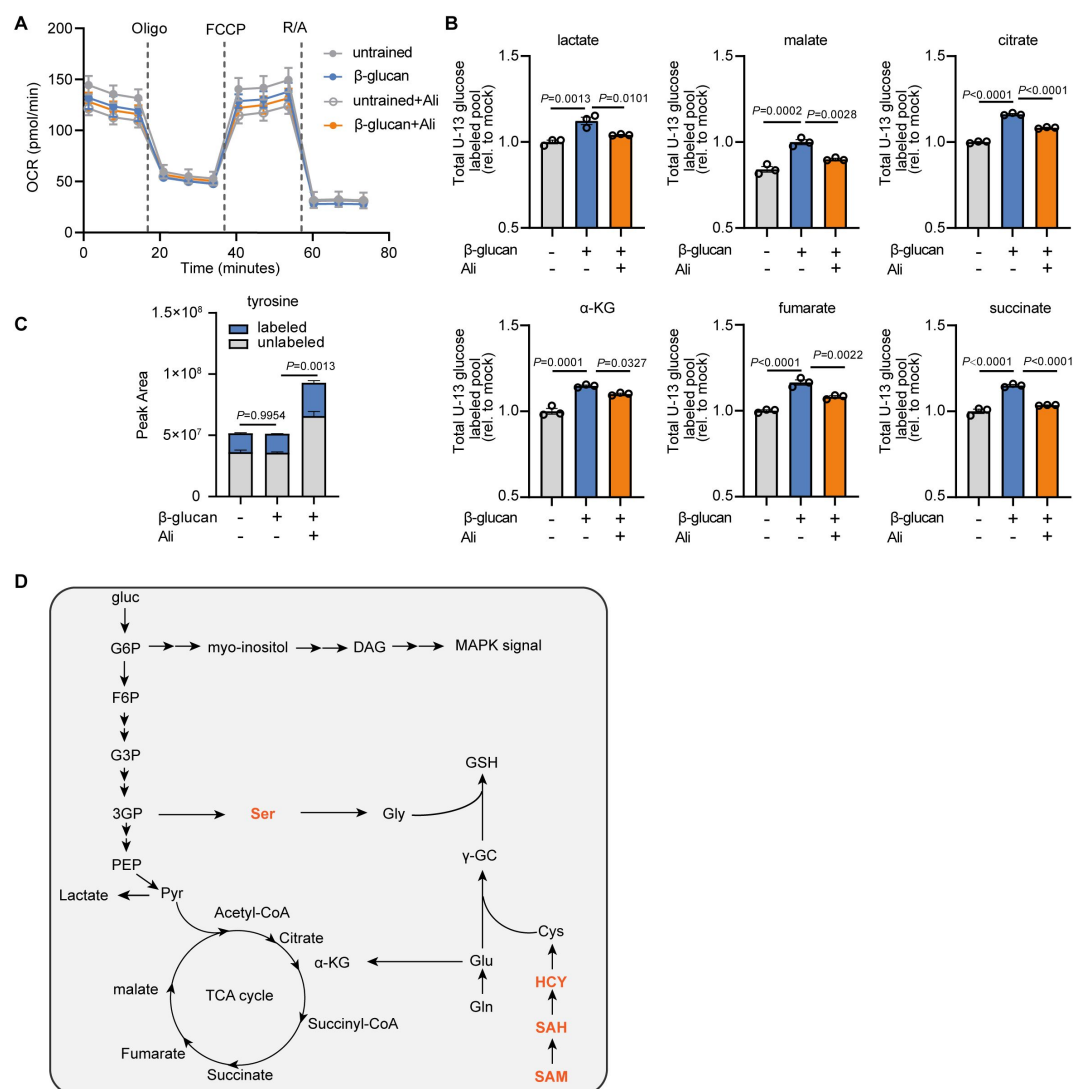


Figure 3—figure supplement 1.

Alisertib inhibits glucose incorporation into glycolysis and TCA cycle.

(A) Oxygen consumption rate (OCR) in BMDMs with different treatment after a mito stress test upon sequential addition of oligomycin (oligo, 1 μ M), FCCP (1 mM), and rotenone/antimycin A (0.5 μ M), as indicated; **(B)** Mass labelling of trained BMDMs with U- 13 C-glucose in the absence or presence of alisertib for 24 h. Training by β -glucan increased incorporation of 13 C-glucose into glycolysis and TCA cycle intermediates; this was reversed by alisertib. **(C)** Peak area of tyrosine and comparing of the sum of peak areas for unlabeled and labeled tyrosine between different treatment groups. **(D)** Diagram illustrating the cross link between glycolysis, TCA cycle, glutathione and SAM. N=3 per group. Data are presented as the mean $\pm$ SEM. and P values were derived from one-way ANOVA (B) or two-way ANOVA (C) with Dunnett's multiple-comparison test. Related to Figure 3.

Figure 4—figure supplement 1.

Inhibition of Aurora kinase A impairs histone trimethylation at H3K4 and H3K36.

Quantification of H3K4me1, H3K9me3, H3K36me3, H3K4me3 and H3K27me3 protein levels was determined by Image Lab software; data are represented as mean $\pm$ SEM of three independent experiments. *P* values were derived from one-way ANOVA with Dunnett's multiple-comparison test compared with β -glucan stimulation only group. Related to [Figure 4](#).

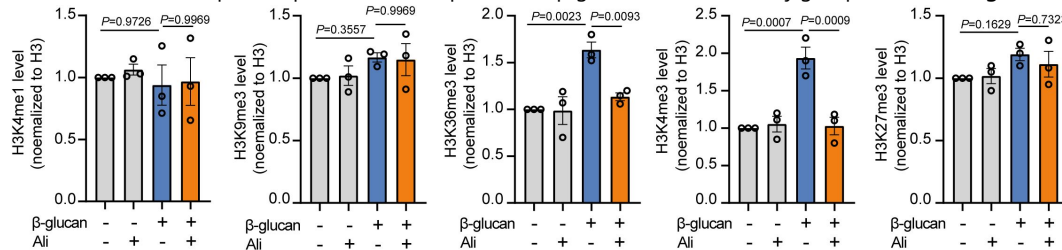
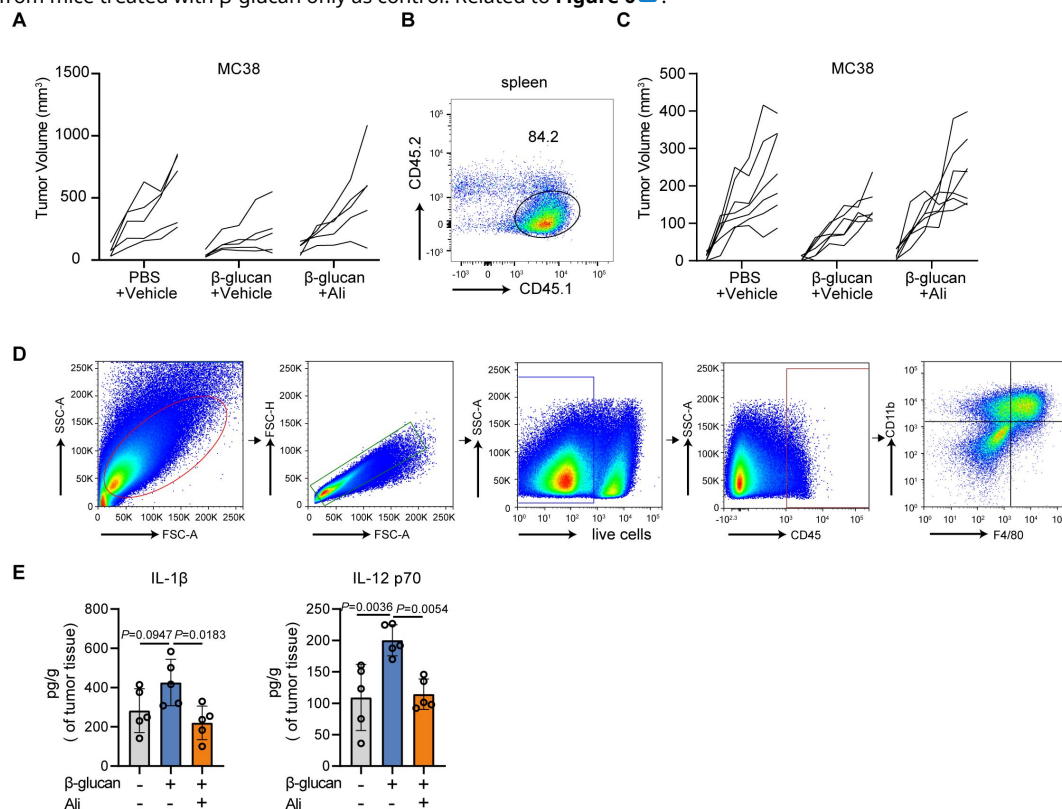


Figure 6—figure supplement 1.

Alisertib abrogates the anti-tumor effect induced by trained immunity.

(A) Individual tumor growth curves ($n=5$ per group). (B) Representative FACS analysis for CD45.1⁺ and CD45.2⁺ cells in the spleen of a chimeric mouse at the end of experiment (day 19 after tumor inoculation). (C) Individual tumor growth curves of chimeric mice ($n=7$ per group). (D) The gating strategy used for analyzing tumor-infiltrated myeloid cells and macrophages. (E) Tumor tissue from [Figure 6A](#) was lysed and the lysates were collected for detection of IL-1 β and IL-12p70 ($n=5$). Data are represented as mean $\pm$ SD. *P* values were derived from one-way ANOVA with Dunnett's multiple-comparison test with data from mice treated with β -glucan only as control. Related to [Figure 6](#).



Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request. Figure source data files contain the numerical data used to generate the figures.

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Author contributions

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Zining Wang: Resources, Writing-review & editing, Funding acquisition

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Ethics

All animal experiments and procedures were reviewed and approved by the Institutional Animal Care and Use Committee of Sun Yat-sen University (2024002174).

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Additional files

Appendix 1. List of compounds in the epigenetic drug library. [↗](#)

Appendix 2. Statistical sources data. [↗](#)

Figure 1–source data 1. *Uncropped and labeled blots for Figure 1D and Figure 1E.* [↗](#)

Figure 1–source data 2. *Raw unedited blots for Figure 1D and Figure 1E.* [↗](#)

Figure 3–source data 1. *Uncropped and labeled blots for Figure 3H and Figure 3I.* [↗](#)

Figure 3–source data 2. *Raw unedited blots for Figure 3H and Figure 3I.* [↗](#)

Figure 4–source data 1. *Uncropped and labeled blots for Figure 4A and Figure 4C.* [↗](#)

Figure 4–source data 2. *Raw unedited blots for Figure 4A and Figure 4C.* [↗](#)

Figure 5–source data 1. *Uncropped and labeled blots for Figure 5A-C, E and G.* [↗](#)

Figure 5–source data 2. *Raw unedited blots for Figure 5A-C, E and G.* [↗](#)

Figure 5–source data 3. *Original images for Figure 5F.* [↗](#)

Figure 6–source data 1. *Intensity analysis for Figure 6F.* [↗](#)

Note

This reviewed preprint has been updated to remove a duplicated heading.

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Reviewer #1 (Public review):

In this updated and improved manuscript, the authors investigate the role of Aurora Kinase A (AurA) in trained immunity, following a broader drug screening aimed at finding inhibitors of training. They show AurA is important for trained immunity by looking at the different aspects and layers of training using broad omics screening, followed up by a more detailed investigation of specific mechanisms. The authors finalised the investigation with an in vivo MC-38 cancer model where AurA inhibition reduces beta-glucan's antitumour effects.

Strengths:

The experimental methods are generally well-described. I appreciate the authors' broad approach to studying different key aspects of trained immunity (from comprehensive transcriptome/chromatin accessibility measurements to detailed mechanistic experiments). Approaching the hypothesis from many different angles inspires confidence in the results.

Furthermore, the large drug-screening panel is a valuable tool as these drugs are readily available for translational drug-repurposing research.

In response to the rebuttal, I would like to compliment and thank the authors for the large amount of work they have done to improve this manuscript. They have removed most of my previous concerns and confusions, and explained some of their approaches in a way that I now agree with them - a great learning opportunity for me as well.

Weaknesses:

(1) The authors have adequately responded to my comments and updated the manuscript accordingly.

(2) The authors have removed most of my concerns. Regarding the use of unpaired tests because that is what is often done in the literature: I still don't agree with this, nor do I think that 'common practice' is a solid argument to justify the approach. However, we can agree to disagree, as I know indeed that many people argue over when paired tests are appropriate in these types of experiments. I appreciate that $n=2$ for sequencing experiments is justifiable in the way these analyses are used as exploratory screening methods with later experimental validation. I also want to thank the authors for reporting biological replicates where relevant and (I should have mentioned this in my original review also) I appreciate they validate some findings in a separate cell line - many papers neglect this important step.

(3) The authors have adequately responded to my comments and updated the manuscript accordingly.

(4) The authors have adequately responded to my comments and updated the manuscript accordingly.

(5) The authors have adequately responded to my comments and updated the manuscript accordingly.

(6) The authors have adequately responded to my comments and updated the manuscript accordingly. They have actually gone above and beyond.

(7) I would like to thank the authors for highlighting this information and taking away my confusion. The authors have adequately responded to my comments and updated the manuscript accordingly.

(8) The authors have adequately responded to my comments and updated the manuscript accordingly.

(9) I still think adding the 'alisertib alone' control would be of great added value, but I can see how it is unreasonable to ask the authors to redo those experiments.

(10) The authors have adequately responded to my comments and updated the manuscript accordingly.

(11) The authors have adequately responded to my comments and updated the manuscript accordingly.

(12) I thank the authors for their work to repeat this experiment with my suggestions included. I am convinced by this nice data. I would recommend that the authors put the data from New Figure 4 also in the manuscript as it adds value to the manuscript (unless I just missed it, I don't see it in Figure 6 or the supplement). Not every reader may look at the reviewer comments/rebuttal documents.

<https://doi.org/10.7554/eLife.104138.2.sa2>

Reviewer #2 (Public review):**Summary:**

This manuscript investigates the inhibition of Aurora A and its impact on β -glucan-induced trained immunity via the FOXO3/GNMT pathway. The study demonstrates that inhibition of Aurora A leads to overconsumption of SAM, which subsequently impairs the epigenetic reprogramming of H3K4me3 and H3K36me3, effectively abolishing the training effect.

Strengths:

The authors identify the role of Aurora A through small molecule screening and validation using a variety of molecular and biochemical approaches. Overall, the findings are interesting and shed light on the previously underexplored role of Aurora A in the induction of β -glucan-driven epigenetic change.

Weaknesses:

Given the established role of histone methylations, such as H3K4me3, in trained immunity, it is not surprising that depletion of the methyl donor SAM impairs the training response. Nonetheless, this study provides solid evidence supporting the role of Aurora A in β -glucan-induced trained immunity in murine macrophages. The part of in vivo trained immunity antitumor effect is insufficient to support the final claim as using Alisertib could inhibit Aurora A in other cell types other than myeloid cells.

Revision:

The authors have satisfactorily addressed the majority of my concerns. In particular, the new bone marrow transplantation data convincingly demonstrate that Aurora A inhibition with Alisertib abolishes the β -glucan-trained antitumor effect—an essential finding supporting the manuscript's conclusions.

<https://doi.org/10.7554/eLife.104138.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer#1 (Public review):

This work regards the role of Aurora Kinase A (AurA) in trained immunity. The authors claim that AurA is essential to the induction of trained immunity. The paper starts with a series of experiments showing the effects of suppressing AurA on beta-glucan-trained immunity. This is followed by an account of how AurA inhibition changes the epigenetic and metabolic reprogramming that are characteristic of trained immunity. The authors then zoom in on specific metabolic and epigenetic processes (regulation of S-adenosylmethionine metabolism & histone methylation). Finally, an inhibitor of AurA is used to reduce beta-glucan's anti-tumour effects in a subcutaneous MC-38 model.

Strengths:

With the exception of my confusion around the methods used for relative gene expression measurements, the experimental methods are generally well-described. I appreciate the authors' broad approach to studying different key aspects of trained immunity (from comprehensive transcriptome/chromatin accessibility measurements to detailed mechanistic experiments). Approaching the hypothesis from many different angles inspires confidence in the results (although not completely - see weaknesses

section). Furthermore, the large drug-screening panel is a valuable tool as these drugs are readily available for translational drug-repurposing research.

We thank the reviewer for the positive and encouraging comments.

Weaknesses:

(1) The manuscript contains factual inaccuracies such as:

(a) Intro: the claim that trained cells display a shift from OXPHOS to glycolysis based on the paper by Cheng et al. in 2014; this was later shown to be dependent on the dose of stimulation and actually both glycolysis and OXPHOS are generally upregulated in trained cells (pmid 32320649).

We appreciate the reviewer for pointing out this inaccuracy, and we have revised our statement to ensure accurate and updated description in manuscript. We are aware that trained immunity involves different metabolic pathways, including both glycolysis and oxidative phosphorylation [1, 2]. We also detected Oxygen Consumption Rate (please see response to comment 8 of reviewer#1) but observed no obvious increase of oxygen consumption in trained BMDMs in our experiment setting. As the reviewer pointed out, it might be dependent on the dose of stimulation.

(b) Discussion: Trained immunity was first described as such in 2011, not decades ago.

We are sorry for the inaccurate description, and we have corrected the statement in our revised manuscript as “Although the concept of ‘trained immunity’ has been proposed since 2011, the detailed mechanisms that regulate trained immunity are still not completely understood.”

(2) The authors approach their hypothesis from different angles, which inspires a degree of confidence in the results. However, the statistical methods and reporting are underwhelming.

(a) Graphs depict mean $\pm$ SEM, whereas mean $\pm$ SD is almost always more informative. (b) The use of 1-tailed tests is dubious in this scenario. Furthermore, in many experiments/figures the case could be made that the comparisons should be considered paired (the responses of cells from the same animal are inherently not independent due to their shared genetic background and, up until cell isolation, the same host factors like serum composition/microbiome/systemic inflammation etc). (c) It could be explained a little more clearly how multiple testing correction was done and why specific tests were chosen in each instance.

We sincerely thank the reviewer for this thoughtful comment. (a) The data from animal experiments in which trained immunity was induced *in vivo* are presented as mean $\pm$ SD, while the statistical results from cell-based experiments are presented as mean $\pm$ SEM in the revised manuscript. (b) We have replaced one-tailed test with two-tailed test (see Figure 3J in revised manuscript, with updated *P* value label). We agree that cells derived from the same animal and subjected to different treatment conditions may be deemed paired data. We reanalyzed our data using paired statistical tests. While this led to a slight reduction in statistical significance for some comparisons, the overall trends remained consistent, and our biological interpretation remains unchanged. For *in vitro* experiments unpaired statistical tests are commonly used in literature [3, 4]. Thus, we still used unpaired test results here. (c) We have provided a detailed description of how multiple comparisons were performed in revised figure legends.

*(d) Most experiments are done with $n = 3$, some experiments are done with $n = 5$. This is not a lot. While I don't think power analyses should be required for simple *in vitro* experiments, I would be wary of drawing conclusions based on $n = 3$. It is also not indicated if the data points were acquired in independent experiments. ATAC-seq/RNA-seq was, judging by the figures, done on only 2 mice per group. No power calculations were done for the *in vivo* tumor model.*

We are sorry for the confusion in our description in figure legends. For the *in vivo* experiment, we determined the sample size ($n=5$, n refers to number of mice used as biological replicates) by referring to the animal numbers used for similar experiments in literatures. And according to a reported resource equation approach for calculating sample size in animal studies [5], $n=5-7$ is suitable for most of our mouse experiments. The *in vitro* cell assay was performed at least three independent experiments (BMs isolated from different mice), and each experiment was independently replicated at least three times and points represents biological replicates in our revised manuscript. In Figure 1A, 5 biological replicates of these experiments are presented to carefully determine a working concentration of alisertib that would not significantly affect the viability of trained macrophages, and that was subsequently used in all related cell-based experiments. As for seq data, we acknowledge the reviewer's concern regarding the small sample size ($n=2$) in our RNA-seq/ATAC-seq experiment. We consider the sequencing experiment mainly as an exploratory/screening approach, and performed rigorous quality control and normalization of the sequencing data to ensure the reliability of our findings. For RNA-seq data analysis, we referred to the DESeq2 manual, which specifies that its statistical framework is based on the Negative Binomial Distribution and is capable of robustly inferring differential gene expression with a minimum of two replicates per group. Therefore, the inclusion of two replicates per group was deemed sufficient for our analysis. Nevertheless, the genomic and transcriptome sequencing data were used primarily for preliminary screening, where the candidates have been extensively validated through additional experiments. For example, we conducted ChIP followed by qPCR for detecting active histone modification enrichment in *Il6* and *Tnf* region to further verify the increased accessibility of trained immunity-induced inflammatory genes.

(e) Furthermore, the data spread in many experiments (particularly BMDM experiments) is extremely small. I wonder if these are true biological replicates, meaning each point represents BMDMs from a different animal? (disclaimer: I work with human materials where the spread is of course always much larger than in animal experiments, so I might be misjudging this.).

Thanks for your comments. In our initially submitted manuscript, some of the statistical results were presented as the representative data (technical replicates) from one of three independent biological replicates (including BMDMs experiments showing the suppression and rescue experiments of trained immunity under different inhibitors or activators, see original Figure 1B-C, Figure 5D, and Figure 5H, also related to Figure 1B-C, Figure 5D, and Figure 5H respectively in our revised manuscript) while other experimental data are biological replicates including CCK8 experiment, metabolic assay and ChIP-qPCR. In response to your valuable suggestion, we have revised the manuscript to present all statistical results as biological replicates from three independent experiments (presented as mean $\pm$ SEM), and we have provided all the original data for the statistical analysis results (please see Appendix 2 in resubmit system).

(3) Maybe the authors are reserving this for a separate paper, but it would be fantastic if the authors would report the outcomes of the entire drug screening instead of only a selected few. The field would benefit from this as it would save needless repeat

experiments. The list of drugs contains several known inhibitors of training (e.g. mTOR inhibitors) so there must have been more 'hits' than the reported 8 Aurora inhibitors.

Thank you for your suggestion and we have briefly reported the outcomes of the entire drug screening in the revised manuscript. The targets of our epigenetic drug library are primarily categorized into several major classes, including Aurora kinase family, histone methyltransferase and demethylase (HMTs and KDMs), acetyltransferase and deacetylase (HDACs and SIRT6s), JAK-STAT kinase family, AKT/mTOR/HIF, PARP family, and BRD family (see New Figure 1, related to Figure 1-figure supplement 1B in revised manuscript). Notably, previous studies have reported that inhibition of mTOR-HIF1 α signaling axis suppressed trained immunity [6]. Our screening results also indicated that most inhibitors targeting mTOR-HIF1 α signaling exhibit an inhibitory effect on trained immunity. Additionally, cyproheptadine, a specific inhibitor for SETD7, which was required for trained immunity as previously reported [7], was also identified in our screening.

JAK-STAT signaling is closely linked to the interferon signaling pathway, and certain JAK kinase inhibitors also target SYK and TYK kinases. A previous drug library screening study has reported that SYK inhibitors suppressed trained immunity [8]. Consistently, our screening results reveal that most JAK kinase inhibitors exhibit suppressive effects on trained immunity.

BRD (Bromodomain) and Aurora are well-established kinase families in the field of oncology. Compared to BRD, the clinical applications of the Aurora kinase inhibitor are still at early stage. In previous studies using inflammatory arthritis models where trained immunity was established, both adaptive and innate immune cells exhibited upregulated expression of AurA [9, 10]. Our study provides further evidence supporting an essential role of AurA in trained immunity, showing that AurA inhibition leads to the suppression of trained immunity.

(4) Relating to the drug screen and subsequent experiments: it is unclear to me in supplementary figure 1B which concentrations belong to secondary screens #1/#2 - the methods mention 5 μ M for the primary screen and "0.2 and 1 μ M" for secondary screens, is it in this order or in order of descending concentration?

Thank you for your comments and we are sorry for unclear labelled results in original manuscript (related to Figure 1-supplement 1C). We performed secondary drug screen at two concentrations, and drug concentrations corresponding to secondary screen#1 and #2 are 0.2 and 1 μ M respectively. It was just in this order, but not in an order of descending concentration.

(a) It is unclear if the drug screen was performed with technical replicates or not - the supplementary figure 1B suggests no replicates and quite a large spread (in some cases lower concentration works better?)

Thank you for your question. The drug screen was performed without technical replicates for initial screening purpose, and we need to verify any hit in the following experiment individually. Yes, we observed that lower concentration works better in some cases. We speculate that it might be due to the fact that the drug's effect correlates positively with its concentration only within a specific range. But in our primary screening, we simply choose one concentration for all the drugs. This is a limitation for our screening, and we acknowledge this limitation in our discussion part.

(5) The methods for (presumably) qPCR for measuring gene expression in Figure 1C are missing. Which reference gene was used and is this a suitably stable gene?

We are sorry for this omission. The mRNA expression of *Il6* and *Tnf* in trained BMDMs was analyzed by a quantitative real-time PCR via a DDCT method, and the result was normalized to untrained BMDMs with *Actb* (β -actin) as a reference gene, a well-documented gene with stable expression in macrophages. We have supplemented the description for measuring gene expression in Material and Methods in our revised manuscript.

(6) From the complete unedited blot image of Figure 1D it appears that the p-Aurora and total Aurora are not from the same gel (discordant number of lanes and positioning). This could be alright if there are no/only slight technical errors, but I find it misleading as it is presented as if the actin (loading control to account for aforementioned technical errors!) counts for the entire figure.

We are very sorry for this omission. In the original data, p-Aurora and total Aurora were from different gels. In this experiment the membrane stripping/reprobing after p-Aurora antibody did not work well, so we couldn't get all results from one gel, and we had to run another gel using the same samples to blot with anti-aurora antibody and used β -tubulin as loading control for total AurA (please see New Figure 2A, also related to original Figure 1D). We have provided the source data for β -tubulin from the same membrane of total AurA (please see Figure 1-source data). To avoid any potential misleading, we have repeated this experiment and updated this Figure (please see New Figure 2B, also related to Figure 1D in revised manuscript) with phospho-AurA, total AurA and β -actin from the same gel. The bands for phospho AurA (T288) were obtained using a new antibody (Invitrogen, 44-1210G) and we have revised this information in Material and Methods. We have provided data of three biological replicates to confirm the experiment result also see New Figure 2B, related to Figure 1D in revised manuscript, and the raw data have been added in source data for Figure 1)

(7) Figure 2: This figure highlights results that are by far not the strongest ones - I think the 'top hits' deserve some more glory. A small explanation on why the highlighted results were selected would have been fitting.

We appreciate the valuable suggestion. Figure 2 (see also Figure 2 in revised manuscript) presented information on the chromatin landscape affected by AurA inhibition to confirm that AurA inhibition impaired key gene activation involved in pro-inflammatory macrophage activation by β -glucan. In Figure 2B we highlighted a few classical GO terms downregulated including "regulation of growth", "myeloid leukocyte activation" and "MAPK cascade" (see also Figure 2B in revised manuscript), among which "regulation of growth" is known function of Aurora A, just to show that alisertib indeed inhibited Aurora A function in vivo as expected. "Myeloid leukocyte activation" and "MAPK cascade" were to show the impaired pro-inflammatory gene accessibility. We highlighted KEGG terms downregulated like "JAK-STAT signaling pathway", "TNF signaling pathway" and "NF-kappa B signaling pathway" in Figure 2F (see also Figure 2F in revised manuscript), as these pathways are highly relevant to trained immunity. Meanwhile, KEGG terms "FOXO signaling pathway" (see also Figure 2G in revised manuscript) was highlighted to confirm the anti-inflammation effect of alisertib in trained BMDMs, which was further illustrated in Figure 5 (see also Figure 5 in revised manuscript, illustrating FOXO3 acts downstream of AurA). Some top hits in Figure 2B like "positive regulation of cell adhesion", and "pathway of neurodegeneration" and "ubiquitin mediated proteolysis" in Figure 2F and 2G, is not directly related to trained immunity, thus we did not highlight them, but may provide some potential information for future investigation on other functions of Aurora A.

(8) Figure 3 incl supplement: the carbon tracing experiments show more glucose-carbon going into TCA cycle (suggesting upregulated oxidative metabolism), but no mito stress

test was performed on the Seahorse.

We appreciate this question raised by the reviewer. We previously performed Seahorse XF analyze to measure oxygen consumption rate (OCR) in β -glucan-trained BMDMs. The results showed no obvious increase in oxidative phosphorylation (OXPHOS) indicated by OCR under β -glucan stimulation (related to Figure 3-figure supplement 1 A) although the carbon tracing experiments showed more glucose-carbon going into TCA cycle. We speculate that the observed discrepancy between increased glucose incorporation into TCA cycle and unchanged OXPHOS may reflect a characteristic metabolic reprogramming induced by trained immunity. The increased incorporation of glucose-derived carbon into the TCA cycle likely serves a biosynthetic purpose—supplying intermediates for anabolic processes—rather than augmenting mitochondrial respiration[6]. Moreover, the unchanged OXPHOS may be attributed to a reduced reliance on fatty acid oxidation- “catabolism”, with glucose-derived acetyl-CoA becoming the predominant substrate. Thus, while overall OXPHOS remains stable, the glucose contribution to the TCA cycle increases. This is in line with reports showing that trained immunity promotes fatty acid synthesis- “anabolism”[11]. Alternatively, the partial decoupling of the TCA cycle from OXPHOS could result from the diversion of intermediates such as fumarate out of the cycle. Oxygen consumption rate (OCR) after a mito stress test upon sequential addition of oligomycin (Oligo, 1 μ M), FCCP (1 mM), and Rotenone/antimycin (R/A, 0.5 μ M), in BMDMs with different treatment for 24 h. β -glucan, 50 μ g/mL; alisertib, 1 μ M.

(9) Inconsistent use of an 'alisertib-alone' control in addition to 'medium', 'b-glucan', 'b-glucan + alisertib'. This control would be of great added value in many cases, in my opinion.

Thank you for your comment. We appreciate that including “alisertib-alone” group throughout all the experiments may further solidify the results. We set the aim of the current study to investigate the role of Aurora kinase A in trained immunity. Therefore, in most settings, we did not include the group of alisertib only without β -glucan stimulation.

(10) Figure 4A: looking at the unedited blot images, the blot for H3K36me3 appears in its original orientation, whereas other images appear horizontally mirrored. Please note, I don't think there is any malicious intent but this is quite sloppy and the authors should explain why/how this happened (are they different gels and the loading sequence was reversed?)

Thank you for pointing out this error. After checking the original data, we found that we indeed misassembled the orientation of several blots in original data submitted. We went through the assembling process and figured out that the orientation of blots in original data was assembled according to the loading sequences, but not saved correctly, so that the orientations in Figure 4A were not consistent with the unedited blot image. We are sorry for this careless mistake, and we have double checked to make sure all the blots are correctly assembled in the revised manuscript. We also provided three replicates of for the Western blot results showing the level of H3K36me3 in trained BMDMs was inhibited by alisertib (as seen in New Figure 7 at recommendation 2 of reviewer#2).

(11) For many figures, for example prominently figure 5, the text describes 'beta-glucan training' whereas the figures actually depict acute stimulation with beta-glucan. While this is partially a semantic issue (technically, the stimulation is 'the training-phase' of the experiment), this could confuse the reader.

Thanks for the reviewer's suggestion and we have reorganized our language to ensure clarity and avoid any inconsistencies that might lead to misunderstanding.

(12) Figure 6: Cytokines, especially IL-6 and IL-1 β , can be excreted by tumour cells and have pro-tumoral functions. This is not likely in the context of the other results in this case, but since there is flow cytometry data from the tumour material it would have been nice to see also intracellular cytokine staining to pinpoint the source of these cytokines.

Thanks for the reviewer's suggestion. In Figure 6, we performed assay in mouse tumor model and found that trained immunity upregulated cytokines level like IL-6 in tumor tissue, which was downregulated by alisertib administration. In order to rule out the possibility that the detected cytokines such as IL-6 was from tumor cells, we performed intracellular cytokine staining of single cells isolated from tumor tissues (please see New Figure 4). The result showed that only a small fraction of non-immune cells (CD45⁻ population) expressed IL-6 (0.37% $\pm$ 0.11%), whereas a significantly higher proportion of IL-6-positive cells was observed among CD45⁺ population (deemed as immune cells, 13.66% $\pm$ 1.82%), myeloid cells (CD45⁺CD11b⁺, 15.60% $\pm$ 2.19%), and in particular, macrophages (CD45⁺CD11b⁺F4/80⁺ 37.24% $\pm$ 3.04%). These findings strongly suggest that immune cells, especially macrophages, are the predominant source of IL-6 cytokine within the tumor microenvironment. Moreover, we also detected higher IL-6 positive population in myeloid cells and macrophages (please see Figure 6I in revised manuscript).

Reviewer#2 (Public review):

Summary:

This manuscript investigates the inhibition of Aurora A and its impact on β -glucan-induced trained immunity via the FOXO3/GNMT pathway. The study demonstrates that inhibition of Aurora A leads to overconsumption of SAM, which subsequently impairs the epigenetic reprogramming of H3K4me3 and H3K36me3, effectively abolishing the training effect.

Strengths:

The authors identify the role of Aurora A through small molecule screening and validation using a variety of molecular and biochemical approaches. Overall, the findings are interesting and shed light on the previously underexplored role of Aurora A in the induction of β -glucan-driven epigenetic change.

We thank the reviewer for the positive and encouraging comments.

Weaknesses:

Given the established role of histone methylations, such as H3K4me3, in trained immunity, it is not surprising that depletion of the methyl donor SAM impairs the training response. Nonetheless, this study provides solid evidence supporting the role of Aurora A in β -glucan-induced trained immunity in murine macrophages. The part of in vivo trained immunity antitumor effect is insufficient to support the final claim as using Alisertib could inhibits Aurora A other cell types other than myeloid cells.

We appreciate the question raised by the reviewer. Though SAM generally acts as a methyl donor, whether the epigenetic reprogram in trained immunity is directly linked to SAM metabolism was not formally tested previously. In our study, we provided evidence suggesting the necessity of SAM maintenance in supporting trained immunity. As for in vivo tumor model, we agree that alisertib may inhibits Aurora A in many cell types besides myeloid cells. To further address the reviewer's concern, we have performed the suggested bone marrow transplantation experiment (trained mice as donor and naïve mice as recipient) to verify the contribution of myeloid cell-mediated trained immunity for antitumor

effect (please see New Figure 8, also related to Figure 6C, 6D and Figure 6-figure supplement 1B and 1C in revised manuscript).

Reviewer #1 (Recommendations for the authors):

Some examples of spelling errors and other mistakes (by far not a complete list):

*(a) Introduction, second sentence: reads as if *Candida albicans* (which should be italicised and capitalised properly) and BCG are microbial polysaccharide components.*

(b) Methods: ECAR is ExtraCellular Acidification Rate, not 'Extracellular Acid Ratio'

(c) Figure 2C: β -glucan is misspelled in the graph title.

(d) TNFa has been renamed to 'TNF' for a long time now.

(e) Inconsistent use of Tnf and Tfn (the correct gene symbol is Tnf) (NB: this field does not allow me to italicise gene symbols)

(f) Figure supplement 1B: 'secdonary'

(g) Caption of figure 4: "Turkey's multiple-comparison test"

(h) etc

I would ask the authors that they please go over the entire manuscript very carefully to correct such errors.

We apologize for these errors and careless mistakes. We greatly appreciate your suggestions, and have carefully proofread the revised manuscript to make sure no further mistakes.

Please also address the points I raised in the public review about statistical approaches. Even more important than the relatively low 'n' is my question about biological replicates. Please clarify what you mean by 'biological replicate'. If you are able to repeat at least the in vitro experiments (if this is too much work pick the most important ones) a few more times this would really strengthen the results.

Thank you for your comment. Our biological replicates refer to independently repeated experiments using bone marrow cells isolated from different mice, and n represents the number of mice used. We repeated each experiment at least three times using BMDMs isolated from different mice (n =3, biological replicates). Specifically, we repeated several in vitro experiments showing inhibition of AurA upregulated GNMT in trained BMDMs and showing transcription factor FOXO3 acted as a key protein in AurA-mediated GNMT expression to control trained immunity as well as showing mTOR agonist rescued trained immunity inhibited by alisertib (see New Figure 5, related to Figure 5B-C, Figure 5H in revised manuscript). Additionally, we have provided data with three biological replicates to show the β -glucan induced phosphorylation of AurA (see comment 6 of reviewer#1) and changes of histone modification marker under AurA inhibition and GNMT deficiency (see recommendation 2 of reviewer#2). We also repeated in vivo tumor model to analysis intratumor cytokines (see recommendation 12 of reviewer#1).

Finally: the authors report 'no funders' during submission, but the manuscript contains funding details. Please modify this in the eLife submission system if possible.

Thank you for your kind reminder and we have modified funding information in the submission system.

Reviewer #2 (Recommendations for the authors):

(1) I have the following methodological and interpretative comments for consideration:

Aurora A has been previously implicated in M1 macrophage differentiation and NF- κ B signaling. What is the effect of Aurora A inhibition on basal LPS stimulation? Considering that β -glucan + Ali also skews macrophage priming towards an M2 phenotype, as shown in Fig. 2E, further clarification on this point would strengthen the study.

Thanks for your suggestion. Previous study showed AurA was upregulated in LPS-stimulated macrophages and the inhibition of AurA downregulated M1 markers of LPS-stimulated macrophages through NF- κ B pathway but did not affect IL-4-induced M2 macrophage polarization [12]. Consistently, we also found that AurA inhibition downregulated inflammatory response upon basal LPS stimulation as shown by decreased IL-6 level (see New Figure 6). In original Figure 2E (also related to Figure 2E in revised manuscript), we showed an increased accessibility of Mrc1 and Chil3 under “ β -glucan + Ali” before re-challenge, both of which are typical M2 macrophage markers. Motif analysis showed that AurA inhibition would upregulate genes controlled by PPAR γ (STAT6 was not predicted). Different from STAT6, a classical transcriptional factor in controlling M2 polarization (M2a) dependent on IL-4 or IL-13, PPAR γ mediates M2 polarization toward M2c and mainly controls cellular metabolism on anti-inflammation independent on IL-4 or IL-13. Thus, we speculate that inhibition of AurA might promote non-classical M2 polarization, and the details warrant future investigation.

(2) In Figure 4A, it looks like that H3K27me3 is also significantly upregulated by β -glucan and inhibited by Ali. How many biological replicates were performed for these experiments? It would be beneficial to include densitometric analyses to visualize differences across multiple Western blot experiments for better reproducibility and quantitative assessment. In addition, what is the effect of treatment of Ali alone on the epigenetic profiling of macrophages?

We are sorry for this confusion. Each experiment was performed with at least three independent biological replicates. In original Figure 4-figure supplement 1 (also related to Figure 4-figure supplementary 1 in the revised manuscript), we presented the densitometric analysis results from three independent Western blot experiments, which showed that β -glucan did not affect H3K27me3 levels under our experimental conditions. Three biological replicates data for histone modification were shown as follows (New Figure 7, as related to Figure 4-figure supplement 1 in revised manuscript). We appreciate that assay for “Ali alone” in macrophages may add more value to the findings. We set the aim of the current study to investigate the role of Aurora kinase A in trained immunity, and we know that alisertib itself would not induce or suppress trained immunity. Therefore, in most settings, we did not test the effect of Alisertib alone without β -glucan stimulation.

(3) The IL-6 and TNF concentrations exhibit considerable variability (Fig. 3K and Fig. 5H), ranging from below 10 pg/mL to 500-1000 pg/mL. Please specify the number of replicates for these experiments and provide more detail on how variability was managed. Including this information would enhance the robustness of the conclusions.

Thank you for your comment. These experiments were replicated as least three times using BMDMs isolated from different mice. The observed variations in cytokines concentration may be attributed to factors such as differences in cell density, variability among individual mice, and the passage number of the MC38 cells used for supernatant collection. We have prepared new batch of BMDMs and repeated the experiment and provided consistent results in the

revised manuscript (please see Figure 5H in revised manuscript). Data for biological replicates have been provided (please see Appendix 2 in resubmit system).

(4) The impact of Aurora A inhibition on β -glucan-induced anti-tumor responses appears complex. Specifically, GNMT expression is significantly upregulated in F4/80⁻ cells, with stronger effects compared to F4/80⁺ cells as seen in Fig. 6D. To discern whether this is due to the abolishment of trained immunity in myeloid cells or an effect of Ali on tumor cells which inhibit tumor growth, I suggest performing bone marrow transplantation. Transplant naïve or trained donor BM into naïve recipients, followed by MC38 tumor transplantation, to clarify the mechanistic contribution of trained immunity versus off-target effects.

Thanks for your valuable suggestion. Following your suggestion, we have performed bone marrow transplantation to clarify that alisertib acts on the BM cells to inhibit anti-tumor effect induced by trained immunity (see New Figure 8, related to Figure 6C-D in revised manuscript). As the results shown below, transplantation of trained BM cells conferred antitumor activity in recipient mice, while transplantation of trained BM cells with alisertib treatment lost such activity, further demonstrating that alisertib inhibited AurA in trained BM cells to impair their antitumor activity.

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